

Counterparts

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Draft of 1st April 2015
Comments welcome

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and worlds to truth values, instead of just functions from worlds to truth values, as arguments. In this tradition, a sentence like (27) is assigned a logical form along the lines of (32):

(32) John [believes_{t_0} [λt_1 Harvey is asleep $_{t_1}$]]

or perhaps (33):

(33) John [$\text{believes}_{t_0, w_0}$ [$\lambda t_1 \lambda w_1$ Harvey is asleep $_{t_1, w_1}$]].

But given that the belief schema is part of the background propositional role we are taking for granted, our account of propositions is essentially dictated by our account of the kind of argument taken by 'believes'. If this argument is a function from times and worlds to truth values, then propositions must either be identical to, or at least be determined by, such functions. And obviously, if such entities can be true or false at all, they can be temporarily true or false (and contingently true or false). The point is obscured by the fact that semanticists often *stipulate* that 'proposition' means 'function from worlds to truth-values', and characterise the argument as showing that 'believes' does not express a relation between people and propositions. But this way of talking is not an option for us, given our acceptance of the belief schema. And this is not just a verbal question about the proper use of a theoretical term! While the word 'proposition' is not that common in ordinary life, quantification over and reference to propositions is commonplace; unless the belief schema and the T-schema are true for this ordinary notion, we will lack a good account of arguments like 'John believes that Harvey is asleep, and Harvey is asleep, so John believes something true'.

2.6 The primordial objection to the B-theories

At this point I should put my cards on the table and admit that I completely agree with the dominant reaction to the modal B-theory: namely, that it is completely mad. To echo Russell: 'In such theories, it seems to me, there is a failure of that feeling for reality which ought to be preserved even in the most abstract studies'.¹⁷ The modal B-theory sucks all the *reality* out of reality, leaving nothing but a bloodless husk.

If the metaphysical horror of the modal B-theory is not already apparent to you, it might be a helpful exercise to see how the view sounds if we replace the technical term 'world' with another technical term, 'perspective':

¹⁷Russell was talking about Meinongianism, which seems positively level-headed by comparison with the modal B-theory.

The Earth has different shapes relative to different perspectives. From one perspective, it is round. From another perspective, it is flat. When we say ‘the Earth is round’, we mean only that it is round from a certain perspective. From that perspective, we are saying something true. However, there is another equally valid perspective from which we are speaking falsely, and those who say ‘the Earth is flat’ are speaking truly. These perspectives are all metaphysically on a par. There are no facts about which of them are “correct” or “incorrect”, over and above the fact that each one is correct from its own point of view. We can certainly say ‘We who believe that the Earth is round are right, and those who believe that it is flat are wrong’. But when we do this, all we manage to say is that we are right and they are wrong *from a certain perspective*. And while there are some perspectives relative to which we are right and they are wrong, there are also some perspectives relative to which they are right and we are wrong; relative to the former perspectives, we said something true; relative to the latter, we said something different, and false.

This is almost indistinguishable from the sort of speech one might expect from the “postmodern relativists” who feature in analytic philosophers’ nightmares. Granted, there are a couple of differences. First, modal B-theorists allow for *some* propositions—namely, the necessary ones—to be the case at all worlds. They are like selective postmodernists, who respect mathematics while allowing other sciences (namely, those that deal in the contingent) nothing more than perspective-relative success. Second, there is nothing in the modal B-theorists’ ideology corresponding to the notion of a perspective being the perspective “of” a particular person or group of people. We can consider worlds at which the relevant person or group have only true beliefs. But in general, there will be many such worlds, and they will have no other interesting features in common. (At one world where I have true beliefs, the Earth is round and believed by me to be round at that world; at another, it is flat and believed by me to be flat at that world.) However, I do not think that the ideology of perspective-ownership is crucial to the wrongheadedness of “postmodern relativism”. What is misguided about that view is its stance of neutrality as between true and false hypotheses. And (so long as the hypotheses in question are contingent), this is a feature shared by the modal B-theory.

If reality hasn’t been treating you well, the modal B-theory might seem to offer consolation. Did your rival beat you to that desirable promotion?

If the modal B-theory is true, the situation in reality as a whole is perfectly symmetric: at one world, he beats you; at another world, you beat him. Neither world is more “your” world than the other. Admittedly, there may be many worlds w such that at w , you only care how things go at w . Maybe this is true for every w in which your concerns are rational, so that all the worlds in which you find the modal B-theory consoling are worlds where you are irrational. But on the other hand, each of us is irrational at some worlds and rational at others. From the perspective which finds consolation in the modal B-theory, a wish to avoid irrationality at *this* world rather than some other world looks as arbitrary as any other concern that privileges some worlds over others.¹⁸

(Let me emphasise once again that this objection is not directed at Lewis’s view. Figuring out how Lewis’s view fits into our classification is a task for chapters 3 and 4.)

The primordial objection to the modal B-theory is really an objection to modal egalitarianism. Those who combine modal egalitarianism with propositional contingentism by appealing to the ideology of “factuality”, admitting the existence of contingent truths but denying that any of them are *factually* true, seem no better off than propositional necessitists. To the extent that we are entitled to any pre-theoretic beliefs about what is *factually* the case, we are entitled to think that some contingent truths, such as the truth that the Earth is flat, are nevertheless factually true. Questions about the shape of the Earth seem *not at all* like ethical questions as conceived by expressivism. Our understanding of the ‘factually’ operator is tenuous at best; if it emerges that we are not even allowed to treat questions about the shape of the Earth as paradigms of factuality, many of us will want to throw up our hands and conclude that we don’t understand it at all.

¹⁸Can a B-theorist avoid the primordial objection by positing that each ordinary object is associated with a unique “home world”, glossing questions like ‘Is the Earth *really* round or flat?’ as ‘Is the Earth round or flat at its home world?’ I don’t see a stable view here. For one thing, the epistemology is a disaster. For most worlds w , people who at w have certain experiences of a certain object, and on the basis of these experiences come to believe at w that the object has such and such features *at its home world*, will come to believe a lot of falsehoods at w . If someone ends up having true beliefs at w by using such methods, it will just be due to a lucky match between w and the relevant object’s home world. Even at these lucky worlds, bad counterfactuals will be true: ‘If we had discovered life on Mars, and come to believe on that basis that there was life on Mars at Mars’s home world, we would have been wrong’.

Some formulations of counterpart theory suggest a view tantamount to the modal B-theory supplemented with ‘home world’ ideology. But as we will see, there are much worthier views with at least as good a claim to the label ‘counterpart theory’.

Such is the primordial objection to the modal B-theory. Perhaps it is misleading for me to call it an 'objection': certainly I haven't been able to uncover anything with the shape of an argument with discernible premises. Still: philosophy is full of crazy views which we have decisive reason to reject, whether or not we can find any dialectically useful arguments against them; and the modal B-theory is one of those views.

Some philosophers feel about the temporal B-theory very much as I feel about the modal B-theory. To them, the temporal egalitarian's view of the present time as "just one instant among many" seems wrongheaded in the same way as the modal egalitarian's view of the actual world as "just one world among many". According to them, the temporal B-theory sucks all the life out of the world, and replaces it with a mere static simulacrum.

Having been exposed to spacetime diagrams at an impressionable age, I cannot quite work myself into a state of visceral identification with this reaction to the temporal B-theory. But I also cannot dismiss the reaction out of hand. For I can see all too clearly how any remarks I might make to justify such dismissal will be thrown back in my face by modal B-theorists. Modal B-theorists have strong responses to all the articulate arguments against their view, responses which parallel the responses temporal B-theorists can give to the analogous arguments against *their* view. And yet, the modal B-theory is *obviously false*. Could it be that the dialectical situation is similar for the temporal B-theory? The thought cannot be easily brushed aside, especially in view of the number of good philosophers who have felt that the temporal B-theory misses something absolutely essential about the true nature of time and change.

The primordial objection to the temporal B-theory has something in common with an argument that Lewis makes as part of his 'problem of temporary intrinsics' (Lewis 1986b, pp. 203–4). Lewis objects to a certain view on the grounds that it wrongly entails that *straightness* and *bentness* are "relations to times", when in fact they are *properties* which can be instantiated *simpli-citer*. However, the view against which Lewis deploys this objection is not the temporal B-theory but only the combination of the temporal B-theory with something called 'endurantism'. According to Lewis, B-theorists can avoid the objection by rejecting endurantism in favour of "perdurantism", also known as 'the doctrine of temporal parts'. Lewis's explanations of 'en-

durantism' and 'perdurantism' are not as clear as one might wish.¹⁹ But the issue relevant to the present argument is as follows. Perdurantists, while they accept that ordinary material objects are straight and bent only relative to particular times, think that in addition to ordinary objects there are objects which last only an instant, some of which are straight or bent *simpliciter*, and some of which are parts of ordinary objects. Moreover, they analyse '*x* is straight at *t*' as '*x* has an instantaneous temporal part at *t* that is straight *simpliciter*'. By contrast, endurantists as Lewis conceives them have no use at all for expressions like 'straight *simpliciter*'.

The two pages where Lewis makes this argument may have attracted more pages of discussion per word than anything else in analytic philosophy other than Gettier. The main thing I want to do here is just to stress that the 'primordial objection' to the temporal B-theory is not at all the same thing as the 'problem of temporary intrinsics', since it is completely neutral as between the endurantist and perdurantist pictures, whatever exactly those amount to. The primordial objection applies to any view according to which *I* am not sitting *simpliciter*, or according to which *the Earth* is not warming *simpliciter*. It makes no difference whether the target of the objection also posits some *other* things, which are not people or planets or objects of any other familiar kind, and claims that some of *these* things are, for example, sitting *simpliciter*.

That said, let me say a little more about Lewis's argument. Many readers have been misled into thinking that the perdurantist view that Lewis favours involves a completely general account of the semantic function of temporal modifiers like '*at t*', according to which for something to be *F* at an instant *t* is for it to have a temporal part which exists only at *t* and is *F simpliciter*. On such a view, being a child or a philosopher at *t* requires having a temporal part that is a child or philosopher *simpliciter*. Surely this is not Lewis's view. Lewis doesn't think that any instantaneous things are *people*: according to the account in Lewis 1976a, a person is a *maximal* mereological fusion of appropriately related instantaneous objects. And it would be absurd to deny that all children and philosophers are people. Thus, '*To be F at t is to have a temporal part that exists only at t and is F simpliciter*' is true, at best, for a certain special subset of 'temporally local' predicates. (This is presumably why Lewis speaks of 'the problem of temporary intrinsics' rather than 'the

¹⁹Hawthorne (2007) surveys various distinct things that might be at issue between 'endurantists' and 'perdurantists'.

problem of temporary properties'.)²⁰

If one thought that Lewis was arguing for the sweeping view that requires instantaneous temporal parts of children and philosophers to be children and philosophers in their own right, one could conceive of the perdurantist's claim as a contribution to *semantics*. On the envisaged semantic theory, the property of being straight *simpliciter* is the semantic value of the word 'straight', while the semantic value of 'at *t*' might be the function that maps any property *p* to the property of having a temporal part at *t* that has *p*. But on Lewis's actual view, no semantics along these lines is defensible. For it is completely implausible to think that predicates that express "temporary intrinsics" have a different kind of semantic value from predicates that do not, with 'at *t*' playing two different semantic roles depending on which kind of predicate it is attached to. Semantically, 'straight' belongs to the same category as adjectives like 'old', 'busy', and 'available', which do not (on Lewis's view) plausibly apply to instantaneous stages. Thus the view must be conceived not as a claim about the semantics of the ordinary predicate 'straight' but as a contribution to its *metaphysics*—*what it is* to be straight at *t* is to have a temporal part at *t* that is straight *simpliciter*. This is not the sort of claim in which semanticists should take a professional interest, any more than they should take an interest in (say) the best-system theory of lawhood, or physicalist accounts of colour.

Suppose Lewis is right that there is a certain property—call it *S*—such that to be straight at an instant is just to have a temporal part at that instant that has *S*. There is still a question about how *S* could come to be referred to by the English word 'straightness', or expressed by the predicate 'straight *simpliciter*'. Ordinary uses of 'straight' never express *S*, since they attach to names that refer to, or variables that range over, long-lived objects. The semantic value of 'straight' in these ordinary uses is the same sort of entity as the semantic value of 'old'—perhaps a function from objects and times to truth values. We cannot force 'straight' to semantically express *s* just by refusing to supply any time argument. And because of this semantic kinship between 'straight' and 'old', it seems quite implausible to think of 'straightness' as referring to something of a quite different sort from 'oldness'. Of course, Lewis is free to introduce a novel philosophical use for 'straightness' on which it refers to *S* (as opposed to referring to whichever entity stands to 'straight' as oldness stands to 'old'). Similarly, he is free to introduce a novel philosophical use for 'straight *simpliciter*' on which it

²⁰Parsons (2002) makes a similar point. ***discuss Sattig***.

expresses *S*. But if we have to think of claims like ‘straightness is a property, not a relation’ and ‘some things are straight *simpliciter*’ as expressed in a special new philosophical jargon, it is hard to see how we could reasonably regard these claims as intuitive starting points, dialectically appropriate for use as premises in an argument for perdurantism.

Moreover, these reflections increase the plausibility of the thought that whatever legitimate intuitive force Lewis’s premise commands is simply a by-product of the intuitive force behind the primordial objection against the B-theory.²¹ If this thought is right, it is good news for B-theoretic endurantists, since it suggests that so long as they are comfortable facing down the primordial objection, they have nothing more to fear from Lewis’s argument. It is also good news for A-theorists, since it suggests that the many B-theorists who have found Lewis’s argument for perdurantism compelling could be brought, by reflection on what is really driving their intuitions, to find the primordial argument against the B-theory equally compelling.

2.7 The argument from spacetime

The weightiest argument for the B-theory of time is based on a certain picture of the fundamental structure of the world inspired by modern spacetime physics. This picture can be captured in a slogan: *The world is a four-dimensional spacetime manifold*.

To get the argument into focus, we will need to try to capture the content of the manifold picture in more precise terms. In doing this we will have to make various choices, both about the general metaphysical question how, in general, one should understand the enterprise of ‘characterising the fundamental structure of the world’, and about how much specific content to build into this particular account of the fundamental structure of the world. Many of these choices will make little or no difference as far as the argument for the temporal B-theory is concerned. But for now, let’s not worry about making the argument needlessly specific: if we can get a workable argument onto the table, we can consider afterwards how its premises might be weakened.

Here is a first stab at spelling out the manifold picture:

The Manifold Picture: Concrete reality consists, fundamentally speaking, of *spacetime points* and *spacetime regions*. The structure of concrete reality

²¹*cite Hinchliff, Zimmerman*

is fully captured by a four-dimensional *differentiable structure* (a criterion for what counts as a “smooth” co-ordinate system) over the set of spacetime points, together with a short list of ‘physically distinguished’ scalar, vector and/or tensor fields (e.g. the metric field, the electromagnetic field, the mass-density field) on this differentiable structure.

This formulation helps itself to some heavy-duty metaphysical jargon which does not wear its meaning on its sleeve—‘fundamentally speaking’ and the notion of ‘fully capturing structure’. The task of finding a disciplined regimentation of this kind of talk is an important one, which has been a focus for much recent discussion. But for present purposes, what we need to do is to extract from *the manifold picture* some consequences expressed in terms of concepts we have already been working with, notably the notion of metaphysical necessity and that of naturalness.

Here is one important claim which can be plausibly taken to follow from *the manifold picture*:

Manifold Supervenience: All true propositions follow with metaphysical necessity from (i) truths about which spacetime points and regions there are; (ii) truths to the effect that a certain point or region is part of a certain other region; (iii) truths about which co-ordinate systems over the set of spacetime points are smooth; (iv) the truth, for each physically distinguished tensor field and spacetime point, about which tensor at that point is the value of that physically distinguished field; and (v) a “that’s all” truth.

The “that’s all” truth referred to in (v) is supposed to be a proposition which somehow captures the idea that there isn’t any more to the world than those spacetime points and/or regions, and that there isn’t any more to the structure of the world than is captured by truths of kinds (ii) and (iii). Of course, the task of explaining what *that* means raises the very same conceptual difficulties as the task of making *the manifold picture* precise. Nevertheless, modal formulations like *Manifold Supervenience* are useful to work with, since we often don’t need to know exactly how to properly articulate a “that’s all” truth in order to make judgments about what it can accomplish.

(Those who think they understand a non-trivial ‘factually’ operator may deny that *Manifold Supervenience* follows from *the manifold picture*. It would

be natural for them to insist that even if the world is fundamentally just a spacetime manifold, only the *factual* true propositions need to be entailed by the facts about spacetime and its fields, while non-factual true propositions can vary independently. But it's worth noting that for the purposes of establishing certain conclusions with a B-theoretic flavour, the result of weakening *Manifold Supervenience* by replacing 'All true propositions' with 'All factual true propositions' would be just as good as *Manifold Supervenience* itself.)

Another kind of claim that seems to be implicit in the manifold picture involves the notion of naturalness:

Manifold Naturalness: The only natural classes of points and regions are those picked out by simple conditions on the physically distinguished fields.

Manifold Naturalness allows us to say, for example, that the set of points where the electromagnetic field vector is zero is a natural class, and that the set of points where the scalar curvature field is positive is a natural class. It is supposed to rule out the idea that there are some 'natural joints' in spacetime that are completely uninteresting from the standpoint of the physically distinguished fields. This is plausibly part of the kind of 'completeness' that the manifold picture claims on behalf of those fields.

Manifold Naturalness talks about the naturalness of classes rather than that of properties. One might contemplate deriving *Manifold Naturalness* from the claim that the physically distinguished fields are the only instantiated (perfectly?) natural properties. But this claim brings up special problems. If we want to think of the metric tensor (for example) as a property, it would seem to be a property of *tensors at points*. And such tensors are standardly treated as rather elaborate set-theoretic constructions. For example, the metric tensor at a point would normally be identified by a certain function from pairs of functions from points to real numbers to real numbers. The idea that non-mathematical properties of such mathematical entities are perfectly natural seems to embody a rather heavyweight form of Platonism. It also raises worries about arbitrariness, since there are many different constructions which can equally well do the job of a tensor at a point (cf. Benacerraf 1965). For this reason, there is good reason to retreat from the claim that the physically distinguished fields are themselves perfectly natural properties to the less committal claim that the perfectly natural properties and relations, whatever they might be, are such that the facts about their pattern of instantiation can be read off the facts about the differentiable structure and the physically

distinguished fields in some reasonably simple way. This leaves us free to entertain many different ideas about what these perfectly natural properties and relations actually are, and how the facts about the physically distinguished tensor fields supervene on them.

Have we done enough by way of articulating the manifold picture to make it clear why it might seem to require a B-theory of time? Not yet. For we haven't said anything about what the 'physically distinguished' fields *are*, or what kind of pattern they make. For all we have said, it might be that, for example, one of the physically distinguished fields is a 'degree of futurity' scalar field, such that the points where the value of this field is zero constitute a natural three-dimensional slice through spacetime which could play the role of the present time (the infamous 'moving spotlight'). There is an intelligible version of the A-theory of time which accepts *Manifold Supervenience* and *Manifold Naturalness* while positing such a field, and claiming that it is metaphysically necessary that the propositions that *will* be true, or *have been* true, are all and only those that *would* be true in a world where the values of the degree-of-futurity field at each point were decreased, or increased, by a constant amount.²² But the picture we are trying to get on the table is not supposed to contain anything like that! To make this clear, let us understand the view to include the following claim:

Manifold Homogeneity: None of the especially natural classes of spacetime points is a connected, spacelike, three-dimensional spacetime region.

(The notion of a region's being 'spacelike' is one that can be understood in terms of spacetime's geometric structure. In both relativistic and Neo-Newtonian spacetime, the relevant kind of structure can be captured by a physically distinguished metric tensor.)

Even this isn't enough to rule out some other kinds of spacetime-friendly A-theory. Some *growing block theorists* may be willing to accept that reality consists, fundamentally, of a spacetime manifold endowed with certain physical fields. It's just that they think that standard B-theorists are wrong about the *extent* of this manifold: rather than going on forever in the 'futurewards' direction, or ending in a 'Big Crunch' at which some of the physical

²²There is also a closely related but arguably more economical view which makes do with a fundamental property of 'presentness' attaching to spacetime points, defining the degree of futurity of a point as the spatiotemporal distance between it and the hypersurface of present points, multiplied by minus one if the point is *before* this hypersurface. For an able defence of this kind of view, see Deasy forthcoming.

fields increase without limit, the manifold just *cuts off* some fourteen billion years after the Big Bang. There is a consistent version of this view which accepts *Manifold Supervenience*: according to it, it is metaphysically necessary that a proposition was, or will be true just in case it would be true in a possible world similar to the actual world in certain specified respects but where the extent of spacetime is smaller or larger.²³ Similarly, a 'shrinking block' theorist might think of reality as a spacetime manifold truncated in the other direction. And even weirder views are conceivable, such as a view that accepts spacetime points corresponding to the past and the future while denying the existence of a 3d slice corresponding to the present. But none of these shapes is the shape of spacetime according to the picture we are trying to get on the table. According to that picture, the spacetime manifold is "as big as it can be" consistent with the physical fields being mathematically well-behaved. Let us make this explicit by setting down *Manifold Completeness* as part of our official statement of the manifold picture:

Manifold Completeness: There is no way to embed the spacetime manifold with its physically distinguished fields into a larger connected manifold of the same dimensionality on which all the physically distinguished fields are everywhere smooth.

Now that we have filled in some of the details in the manifold picture, let us see what we can do to get the argument from the picture to the temporal B-theory into focus. I can see two broad lines of argument, the first of which targets propositional eternalism, and the second of which targets temporal egalitarianism. The first argument can be spelled out as follows:

The Supervenience Argument:

S1 The manifold picture is true.

²³According to the most common version of the growing block view, spacetime is not a manifold in the strict sense but rather a *manifold with boundary*: the result of stitching together a three-dimensional manifold (the present) with a four-dimensional one (the past). This view is inconsistent with *Manifold Homogeneity*, since the class of boundary points is certainly a natural one. However, I see no good reason why the growing block theorist should not, instead, think of the 'forward edge' of reality as topologically *open* rather than closed. On such a view, all spacetime points are 'past' insofar as it makes sense to apply temporal classifications to spacetime points at all. But that need no more prevent proponents of the view from agreeing that all kinds of things are the case *now* than it need prevent them from agreeing that all kinds of things will be the case *later*.

- S2 If the manifold picture is true, then if any propositions are temporarily true, some propositions about the values of the physically distinguished fields at particular spacetime points are temporarily true.
- S3 If the manifold picture is true, then all true propositions about the values of the physically distinguished fields at particular spacetime points are eternally true.

Conclusion: There are no temporarily true propositions.

The second argument is as follows:

The Homogeneity Argument:

- H1 The manifold picture is true.
- H2 If the manifold picture is true, then instants of time are connected, spacelike, three-dimensional regions of spacetime.
- H3 If the manifold picture is true, then no connected, spacelike, three dimensional region of spacetime has an especially natural unit set.

Conclusion: The unit set of the present time is not especially natural.

Let us continue to postpone the question what reason there might be for believing the manifold picture, and consider what can be said on behalf of the remaining premises of these arguments.

S2 can be motivated by appeal to the claim that metaphysical necessity implies eternity (Dorr and Goodman n.d.). Thus, if a proposition follows from some eternally true propositions with metaphysical necessity, that proposition must itself be eternally true. But according to *Manifold Supervenience* (which we have built into the manifold picture) all true propositions follow with metaphysical necessity from truths about which spacetime points there are, about which co-ordinate systems are smooth, about the values of the physically distinguished fields, and a ‘that’s all’ truth. So, if there are temporarily true propositions, there must be temporarily true propositions in at least one of these categories. But it is hard to see how there could be temporarily true propositions about *which spacetime points or regions there are*, or about the smoothness of co-ordinate systems, without there also being temporarily true propositions about the values of the physical fields at points—what counts as a tensor depends on the underlying differentiable

structure. And even without an account of what the ‘that’s all’ truth amounts to, it is hard to see how *it* could be the one source of temporariness. If all the spacetime points and regions that there *are* are things that exist eternally, it is utterly implausible that there *used* to be, or that there *will* be. And it is even more implausible that there used to be, or will be, more ‘kinds of structure’ in spacetime than there now are (let alone that the truth about how this extra structure worked or will work is somehow metaphysically necessitated by the truth about the structure that there is now). Thus, if there are any temporary truths in the supervenience base provided by *Manifold Supervenience*, at least some of it are surely truths of class (iv), about the values of the physically distinguished fields.

(In the form in which I have stated it, S2 will be rejected by non-factualists who restrict *Manifold Supervenience* to factual truths. For them, the interesting argument in the vicinity of the Supervenience Argument is one in which the quantification in *Manifold Supervenience* and in the conclusion is restricted to factual propositions.)

The motivation for S3 is just that it seems quite hard to think of any plausible answer to the question *which* truths about, e.g., the electromagnetic field are temporary, especially given the constraint that the truths about the future and past configurations of that field must be necessitated by truths of the kinds listed in *Manifold Supervenience*. Of course, if our list of physically distinguished included some kind of ‘degree of futurity’ scalar field, there would be a plausible story about how *its* values on particular spacetime points have been different or will be different, namely that when the degree-of-futurity of a spacetime point p is x , it *will* be the case for every $y < x$, and *has* been the case for every $y > x$, that the degree-of-futurity of p is y . But *Manifold Homogeneity* rules this out.²⁴ If we thought that the manifold was in some way “truncated”, we might be able to figure out a story about how its present structure necessitates truths about its future growth or shrinkage, which would make for change even in the overall configuration of familiar physical fields like the electromagnetic field. But given *Manifold Completeness*, the plausible lines of investigation in this direction are blocked. Note that I am *not* claiming that there is anything *incoherent* about the idea that total configuration of the physical fields over the four-dimensional man-

²⁴Certain physically non-standard accounts of the configuration of the standard physical fields—for instance, a view that posits a special slice through spacetime on which these fields fail to be continuous and take value zero—might open up room for similar stories, but these are also hard to reconcile with *Manifold Homogeneity*.

ifold should change—one can in principle imagine A-theoretic views on which reality consists of a four-dimensional manifold undergoing all kinds of changes. The problem is that given what the manifold picture says about the *actual* configuration of the fields, it is hard to think of a plausible account of how that configuration could necessitate any non-trivial truths about how the configuration has been or will be different.

Let us turn next to the Homogeneity Argument. We guaranteed premise H3 by building *Manifold Homogeneity* into our articulation of the manifold picture, so the action is in premise H2. This premise has been almost universally taken for granted in discussions of time and spacetime.²⁵ This practice is encouraged by standard introductions to the concept of spacetime in physics and philosophy, which emphasise the analogy between the role played by spacetime points (in physical theories that talk about spacetime) and the role played in Newtonian physics by ordered pairs comprising a point of absolute space and an instant of time. But we don't want the label 'spacetime' to carry any argumentative weight: if calling the manifold 'spacetime' forced a certain view about how its parts relate to instants of time, we could just call it something else (for example, we could use Forrest's rather catchy term 'the all-pervading aether' ***cite***) without changing our picture of its structure in any way. Nevertheless, there is a strong *prima facie* case for H2, namely that given the basic elements provided by the manifold picture, it is hard to think of any *other* good candidates to identify with the instants of time. Even if we help ourselves to the resources of set theory or some even richer domain of 'constructed' objects, no *interestingly* different candidates to be identified with instants of time jump out at us. (There are many *uninterestingly* different candidates, since there are many possible set-theoretic constructions which correspond in some natural way with regions of the relevant kind; but these don't help with the argument, since *Manifold Homogeneity* entails that none of *them* are especially natural either.) It won't do, for example, to simply announce that instants of time are real numbers, and that the present instant of time is the real number 0, and leave it at that. For we need our metaphysics of instants to fit together with our metaphysics of ordinary objects in such a way as to yield answers to questions like 'What is it for an object to be spherical at an instant?'. And whatever objects might be, it is obscure what it could mean for one of them to be spherical 'at' a real number.

* * *

²⁵****I suppose some citations would be appropriate here. . . .

Thus, there is a strong *prima facie* case that A-theorists must reject the manifold picture — it is little wonder that the literature tends to run something like the manifold picture (or some mild generalisation thereof) together with the B-theory under the label ‘four-dimensionalism’. Some A-theorists will take a radical stand against the manifold picture, claiming that spacetime physics, while perhaps useful as a predictive tool, is radically misleading as a guide to fundamental ontology. Other A-theorists will attempt to keep as much of the manifold picture as they can, making only the minimum alterations required to block the arguments above. Most obviously, they might reject *Manifold Homogeneity* while keeping all the rest of the picture, expanding the list of ‘physically distinguished’ fields to include something like a “degree of presentness” field, or some other structure which picks out a natural slice through spacetime. Call this the ‘Supplemented Manifold Picture’.

As accounts of the fundamental structure of reality go, the Supplemented Manifold Picture seems rather suspicious. If there are good grounds for positing the extra structure that privileges a special three-dimensional slice, they must be extremely different in character from the grounds that support positing, for example, the structure induced by the electromagnetic field. Considered on its own terms, apart from its putative role in explaining the truth of certain descriptions of the world in non-fundamental terms, the privileging of a special three-dimensional slice detracts markedly from the simplicity and elegance of the whole package. The ordinary physical fields fit together in lovely ways. There are simple laws which characterise their interrelations; and the laws for any one field take account of each of the others. Dropping any one of the physical fields would make for a less simple and unified package, since it would leave us without an explanation for certain patterns in the remaining fields. By contrast, the Supplemented Manifold’s special slice-privileging structure is a ‘nomological dangler’. Mentioning it does not help you to describe the patterns in the other fields. The structure is not something you would be tempted to posit, unless you were convinced by a philosophical theory that told you that some such posit was required in order to provide an adequate supervenience base for the familiar empirical facts.

The proponent of the Supplemented Manifold will respond, of course, that a world that lacked this special structure could not be *our* world. Facts about the location of the privileged slice play a crucial role in the supervenience basis for any non-eternal truth, such as the truth that I am sitting. But

we should be on our guard when philosophers come along with controversial accounts of what it takes for familiar truths to obtain, according to which accommodating these truths requires us to make our overall theory of the fundamental structure of the world more complicated than it would otherwise be. Very often, the right response to philosophers in this dialectical position is that their account of the familiar facts is mistaken, and that on a proper account of these facts, their obtaining requires no special-purpose additions to the fundamental structure of the world. An uncontroversial example: some benighted philosopher might claim that we can *just see* that certain objects are *at rest*, and maintain that in order to account for this obvious fact, we must enrich our picture of the fundamental structure of spacetime to include a standard of absolute motion and rest. We respond that a correct metaphysical account of the obvious facts, informed by a correct semantic theory of ‘at rest’ that recognises its vagueness and context-sensitivity, will show that they can obtain even in a world where spacetime lacks this kind of structure. A more controversial example: some dualists claim that the facts about consciousness show that physics is incomplete as an account of the fundamental structure of the world; in some sequences of cases where physics sees only continuous variation, there are sharp natural boundaries between the cases where the “lights of consciousness” are on and the cases where they are off. Physicalists like me reply that a correct account of the obvious facts about consciousness, informed by a correct semantic theory of ‘conscious’ that recognises its vagueness, shows that no such additional natural structure is required for these facts to obtain.²⁶ Of course, considerations of what is required to provide an adequate supervenience basis for the obvious empirical facts are not *irrelevant* to the investigation of the fundamental structure of the world—physics is an experimental discipline! But a lot of the time, we seem to get on better by looking for accounts of the fundamental structure of the world that look simple and unified taken on their own terms, while keeping a very open mind about what might be required to provide an adequate supervenience base for familiar facts. By these standards, the Supplemented Manifold’s natural slice through spacetime looks

²⁶For good measure, here is an *extremely* controversial example that fits the same pattern. Many philosophers of quantum mechanics (*cite Maudlin*) claim that obvious facts, such as the fact that a certain pointer ended up pointing to a certain number, show that the unitarily evolving wavefunction is incomplete or incorrect as an account of the fundamental structure of the world, and must either be supplemented (with ‘hidden variables’) or revised (with ‘collapses’). Many-worlds theorists like (*cite Wallace*) reply that a correct account of the obvious facts in question shows that they require no such supplementation or revision.

like a unwelcome philosophical imposition.

For the sake of concreteness, I have chosen to focus on a rather specific account of the fundamental structure of the world, which is controversial for many reasons that have nothing to do with the philosophy of time. But many of the details of the manifold picture, and of our ways of making it precise are quite irrelevant to the philosophical dialectic. For the Supervenience Argument, all that mattered about the picture was the fact that there is a certain body of truths in which it is hard to see how there could be temporariness, and which, according to it, necessitate every truth. For the Homogeneity Argument, all that mattered was that the only obvious candidates the picture provides play the role of *instants of time* are entities none of which has an especially natural unit class. There are many ways in which we could modify the picture without affecting these features:

- We could introduce further natural properties and relations on the spacetime points and regions that go beyond the kind of structure that is naturally captured in the language of tensor fields. For example, one could use a binary “belongs to the backwards light cone of” relation among spacetime points to capture the manifold’s geometric structure. This introduces a fundamental asymmetry that is not present when the only relevant structure is the metric tensor field.²⁷
- We could reduce the fundamental ontology, perhaps by eliminating regions and keeping only the points, or by eliminating the points and keeping only some special regions (such as the regions with nonzero volume, or the one-dimensional submanifolds).
- We could, more radically, find some other fundamental ontology in terms of which one can capture the mathematical structure of a differential manifold—perhaps even one that does not involve points or regions at all.²⁸ Also, we could consider various *generalisations* of the concept of a differentiable manifold that have been suggested as possibly of physical interest (**cites**).
- We could expand the fundamental ontology by adding a category of *particles*, each of which stands in a fundamental ‘occupation’ relation

²⁷**cite Maudlin

²⁸See Geroch n.d., Dorr 2011b, Arntzenius and Dorr 2011.

to a certain large collection of points (and/or to the corresponding region), in such a way that the resulting pattern still does not provide any natural way of singling out a three-dimensional subregion. This dualistic ontology of spacetime and particles is one that some metaphysicians may find more familiar than one on which the parts of spacetime are the only fundamental entities. But contrary to the impression one might get from many popularisations, modern ‘particle physics’ does very little to support the claim that particles belong on our list of fundamental entities.²⁹

- More generally, we could expand the fundamental ontology by adding a category of ‘occupiers’, with a fundamental ‘occupation’ relation setting up a one-to-one correspondence between occupiers and spacetime regions, or between occupiers and those spacetime regions at which the non-geometric fields are nonzero. To make sure that the occupiers aren’t mere dangles in the arrangement, one might also want to say that some of the fundamental properties and relations that determine the facts about the field-values at a given point are properties of the occupiers, or relations some of whose relata are occupiers.³⁰ It is hard for me to see what could motivate adding occupiers to the fundamental ontology: they merely make the ontology more complex without bringing any obvious theoretical benefits. Nevertheless, provided that the occupiers are sprinkled around spacetime rather than confined to a single special slice, their addition does not interfere with the Homogeneity Argument. And insofar as hard to see how the facts about the occupation relation that establish the one-one mapping from occupiers onto regions could be temporary, the addition of occupiers also leaves the Supervenience Argument standing.

Even allowing for modifications of these sorts, it is doubtful whether the manifold picture is a true account of the actual fundamental structure of the world. The empirical success of quantum theory seems to require some kind of far-reaching revisions to the kinds of hypotheses about fundamental structure I have been considering, and there is almost nothing I could say that would be uncontroversial about what the resulting picture should look like. My own sense is that it is quite unlikely that the evidence that requires us to

²⁹***Cite Wallace (and others?) on the conditions for the emergence of particle-like behaviour in QFT***

³⁰For a clear example of such dualism, see Heller n.d.

replace classical field theory with quantum field theory, or the development of quantum gravity, will have much bearing on the A-theory/B-theory debate. The quantum-theoretic successors of the manifold picture will be *prima facie* just as inhospitable to the A-theory. They too will admit modifications like the Supplemented Manifold which would remove the inhospitability by adding extra fundamental facts which are naturally thought of as changeable and as marking out a new natural division within the classes of entities that look like natural candidates to be the instants. But such modifications will not look any more natural, or more in keeping with the spirit of the theory, than they do in the case of the manifold picture. If I am right about this, we might as well use the manifold picture in its original form as a proxy for whatever more adequate accounts of the structure of the world might emerge from reflection on up-to-date physics.

So far, I have managed to say nothing about relativity theory. As I have spelled it out, the manifold picture is consistent with there being a uniquely natural ‘foliation’ of spacetime (i.e. a division of it into smooth, non-overlapping three-dimensional slices). Pre-relativistic physics can be interpreted as describing a manifold of this sort—what is called ‘Neo-Newtonian spacetime’.³¹ (‘Neo-’ because, by contrast with Newton’s own account of the metaphysical underpinnings of his physics, it provides no *natural* way to make sense of questions like ‘Is this particle in motion or at rest?’ and ‘How fast is it moving?’.) Even if physics turned out to favour Neo-Newtonian spacetime, the imposition of an extra structure picking out a special *slice* from the privileged foliation as “the present” would look suspicious in the way I have been complaining about. However, *relativistic* manifolds have some further features which make the addition of such structure especially alien and unsimple.

In a generic relativistic manifold, there is no simple way to read off a uniquely natural foliation from the geometric structure. The geometric structure of these manifolds *does*, indeed, determine a natural binary relation over spacetime points which we might think of as having the significance of “beforeness”. But by contrast with the Neo-Newtonian case, the natural “beforeness” relation does not induce a natural foliation on the manifold. In Neo-Newtonian spacetime, the relation *x is not before y and y is not before x* is an equivalence relation; its equivalence classes are the leaves of the

³¹***cite Sklar, Earman***

privileged foliation. In the relativistic case, by contrast, this relation (known as ‘spacelike separation’) fails to be an equivalence relation, since it is not transitive. A point z can be spacelike separated from each of two points x and y , even though x is before y . (This is illustrated in figure ***.)

Relativistic spacetimes have the following further feature: once we are *given* a particular spacelike hypersurface S in the manifold, there is a natural and obvious way of making sense of questions like ‘Is this body in motion or at rest relative to S ?’ and ‘How fast is this body moving relative to S ?’ (assuming that “bodies” are things that somehow single out timelike lines through the manifold, their “worldlines”). For in relativistic geometry, unlike the geometry of Neo-Newtonian spacetime, it makes good sense to ask about the *angle* at which a (smooth) timelike line intersects a (smooth) spacelike hypersurface. In particular, it makes sense to ask whether a timelike line is “perpendicular” to a spacelike hypersurface at the point where they intersect.³² When a body’s worldline intersects S perpendicularly, it is natural to say that it is “at rest” relative to S ; when it intersects S at an angle, the magnitude of this angle will correspond to the particle’s *speed* relative to S . Thus, if we were to supplement the standard kinds of spacetime structure by adding an extra fundamental property, the ‘glow of presentness’, which attaches to a single spacelike hypersurface in the manifold, we would find ourselves willy-nilly rehabilitating those awkward Newtonian questions about “absolute” motion and rest. This seems a step backwards. The general consensus is that it is a great advantage of Neo-Newtonian spacetime over Newton’s metaphysics of absolute space that it allows us to avoid having to posit such explanatorily irrelevant, and epistemologically intractable, matters of fact. Relativistic spacetime without a ‘glow of presentness’ preserves this advantage, as does Neo-Newtonian spacetime even with a glow of presentness superadded. But relativistic spacetime with a glow of presentness does not.

This contrast must not be overstated. In certain *particular* relativistic manifolds—indeed, in certain large classes of such manifolds—there are non-arbitrary mathematical recipes for extracting a foliation of the manifold from the facts about the physically distinguished fields. To take a simple example, consider a geometrically flat (Minkowskian) manifold that carries some mass-bearing field, whose total mass is finite when integrated over

³²Be warned, however, that relativistic “angles” do not work at all like ordinary Euclidean angles: on the most natural way of measuring them, the angles at which timelike lines intersect spacelike hypersurfaces can be arbitrary real numbers. ***Cite Maudlin***

any spacelike hypersurface. In this setting, we can pick out a unique foliation by the condition that relative to each of its leaves, the centre of mass of the universe should be at rest. And in some families of *general*-relativistic spacetimes, we can even find non-arbitrary mathematical recipes for extracting foliations just from the behaviour of the *metric* field, without having to attend to any of the other fields. For example, in 'Big Bang' spacetimes with an initial spacelike singularity, we can assign a number to each point that is the length of the longest path running backwards from it to the singularity; if everything goes well, the sets of points that are all assigned the same number by this function will constitute a foliation of the manifold by smooth spacelike hypersurfaces.³³

These facts mean that we have to be careful in explaining the sense in which these relativistic manifolds provide no "privileged" or "absolute" criteria of motion and rest. In the favourable cases, it is possible to introduce a meaning for 'at rest' without doing anything as *manifestly* arbitrary as saying 'I hereby stipulate that things that are at rest relative to the walls of my lab count as being "at rest"', or 'I hereby stipulate that the trajectory that passes through spacetime points x and y counts as being "at rest"'. But this is also true in Neo-Newtonian spacetime. If the total mass of the universe is finite, there are good questions about whether any given trajectory is in motion or at rest relative to the centre of mass of the universe, and if we wanted to stipulate that 'at rest' should mean 'at rest relative to the centre of mass of the universe', there would be no obvious sense in which we had made an *arbitrary* choice. The widespread consensus that Neo-Newtonian spacetime is preferable to Newton's metaphysics as a setting for pre-relativistic physics need not be based on ignorance of this fact. There really is an epistemologically important difference between the hypothesis that spacetime is Neo-Newtonian and Newton's metaphysics of absolute space; the point about the centre of mass of the universe just shows we have to be cautious in using words like 'arbitrary', 'conventional', 'privileged', and 'natural' in capturing the relevant difference. And the fact remains that the relevant attractive feature of Neo-Newtonian spacetime is one that we give up if we want to supplement a relativistic manifold with the 'glow of presentness'.

³³However, some of these spacetimes admit several different mathematically natural foliations, so that there is still something offputtingly arbitrary and epistemologically problematic about a metaphysical theory that selects one of them as *the* one whose leaves sometimes enjoy the glow of presentness. ***see Belot on CMC time, etc.***

Relativity makes the faults of the Supplemented Manifold theory—its lack of simplicity, its disunified character—worse than they would otherwise be. But my assessment is that the weightiest reasons for resisting theories like the Supplemented Manifold are independent of relativity. They are simply the general reasons for being suspicious whenever philosophers look at some elegant, austere account of the fundamental structure of the world and tell us it *must* be incomplete, on the grounds that it doesn't provide what philosophical reflection has revealed to be the necessary conditions for the obtaining of some familiar facts. Attempts to draw conclusions of that sort from philosophical reflection have a terrible track record. So, modesty is called for. Before rejecting any account of fundamental structure on philosophical grounds of this sort, we had better subject the seemingly convincing philosophical reflections to the highest level of scrutiny, on the lookout for loopholes we might have missed.

2.8 Counterpart theory

We seem to be confronted with a stark choice. We could take spacetime physics seriously as a guide to metaphysics, endorse the B-theory of time, and refuse to be moved by the primordial objection. Or we could be swayed by the primordial objection, and conclude that the standard spacetime picture is incorrect or at least radically incomplete as an account of the fundamental structure of the world. As I have admitted, my own feeling for the force of the primordial objection in the temporal case is wavering and insecure. If there are really only these two options, I'll be going with spacetime metaphysics and the B-theory. On the other hand, I can readily identify with the opposite reaction, since I myself am so completely convinced by the primordial objection to the *modal* B-theory. Given this dialectical situation, it is extremely important to know whether we *really are* confronted with this stark choice. Is there a coherent view that lets us accept the account of the world as a homogeneous four-dimensional manifold, without having to think of everyday claims about the temporary properties of ordinary objects as involving implicit relativity to a time?

Indeed there is. It is called *temporal counterpart theory*, and it has been on the map since 1996, when Ted Sider published a paper called 'All the World's a Stage'. Let us see how it achieves the reconciliation we were looking for.

Temporal counterpart theory is a reductive account of what it is to *have once been*, or to be *going to be*, a given qualitative way. The basic idea can be

expressed as follows:

- (TCT) a. To *have been* *F* is to have a past-counterpart which is *F*
 b. To be *going to be* *F* is to have a future-counterpart which is *F*

Here '*F*' stands for some qualitative predicate like 'tired' or 'sitting' (as opposed to a *de re* predicate like 'married to Hillary Clinton'). 'Past-counterpart' and 'future-counterpart' are placeholders; different temporal counterpart theorists are free to give different accounts of what it is for one thing to be a past-counterpart or a future-counterpart of another.³⁴

Temporal counterpart theory is simply the analogue for tense-operators of the modal counterpart theory originated by Lewis in his 1968 paper, 'Counterpart Theory and Quantified Modal Logic'. The basic idea of modal counterpart theory can be expressed like this:

- (MCT) To be *possibly* *F* is to have a counterpart which is *F*

Here again, '*F*' stands for some qualitative predicate, and 'counterpart' is a placeholder which different modal counterpart theorists are free to fill in in different ways. Indeed, in view of the well-known context-sensitivity of 'possibly', we shouldn't want a once-and-for-all account of what it is for one thing to be a counterpart of another: rather, each admissible interpretation of 'possibly' will require a different interpretation of 'counterpart' to validate MCT.

As we have already noticed, Lewis's view, unlike the standard modal B-theory, does not require sentences like 'Humphrey is never elected president' to be understood as involving implicit reference to a particular world. For Lewis, Humphrey is just the sort of thing that orthodox modal A-theorists would take him to be—the sort of thing concerning which it makes sense to ask whether it is, or is not, ever elected president, *simpliciter* as opposed to relative to a given world. And we need to be able to make unrelativised sense of such predications to make sense of the basic idea of modal counterpart theory, as given by MCT. For the same reason, the basic idea of temporal counterpart theory requires us to think of predicates like 'sitting' as applying, or failing to apply, to things *simpliciter*. It would defeat the purpose of TCT and MCT if we insisted on interpreting 'is *F*' as equivalent to 'is *F* at *t*' or

³⁴In Sider's presentation, it looks like the basic notion is just 'temporal counterpart', with 'past-counterpart' being defined as 'temporal counterpart that is past' and 'future-counterpart' as 'temporal counterpart that is future'. TCT is consistent with, but does not require, this further reduction; later I will argue that it is a mistake.

'is *F* at *w*' for some particular instant *t* or world *w*. Moreover, the things which are sitting *simpliciter* are not just temporal parts of people, but people themselves. Since I have been a sitting person, by (33a) one of my past-counterparts is a sitting person (*simpliciter*).

Sider introduced temporal counterpart theory in the context of defending a certain view of the metaphysics of ordinary objects, according to which such objects are *stages*—non-maximal, three-dimensional regions in four-dimensional spacetime, or "occupants" of such regions. "Perdurantists" like Lewis were already used to claiming that some such entities were straight, heavy, happy, in pain. . . ; Sider's "stage view" then adds that it is these entities, rather than the four-dimensional "worms" favoured by perdurantists, that are people, children, planets, tables. . . . This account of what ordinary objects are has the virtue of showing, in a straightforward way, how the manifold picture is consistent with belief in ordinary objects as things capable of falling under temporally variable predicates *simpliciter*, as required by the temporal counterpart theory. However, temporal counterpart theory does not depend on this or any other account of the nature of ordinary objects. It merely requires that our account of ordinary objects should fit with our accounts of what it is to be straight, heavy, happy, in pain, a person, a child, a planet, a table. . . in such a way as to let us make sense of the question whether an ordinary object is one of these ways *simpliciter*.

Lewis's way of building a worked-out theory of modality around the basic idea expressed in MCT involves quite a lot of talk about certain entities he calls 'possible worlds'. In 1986b, it emerges that Lewis's 'possible worlds' are just what we might call *universes*, or *cosmoi*: in Lewis's final analysis, they are maximal mereological fusions of parts linked by spatiotemporal or "analogically spatiotemporal" relations. I will call things that satisfy this definition 'Lewis-worlds'. Lewis argues that there are many Lewis-worlds, none of which is metaphysically privileged in any relevant way. One of the Lewis-worlds—the one he calls 'the actual world'—contains *us* as parts. But this dubious distinction is not one that makes the unit class of that Lewis-world especially natural, since there is nothing very special about *us* that marks us out as different from the inhabitants of other Lewis-worlds. In his claims about 'worlds', then, Lewis sounds like a modal egalitarian. But this is potentially misleading, because of the disparities between the role that Lewis-worlds play in Lewis's theory and the standard 'world' role as outlined in section 1.3. For example, the standard 'world' role allows us to say that for absolutely every object *x*, *x* is a philosopher iff *x* is a

philosopher at the actualised world. But when we look Lewis's suggestions about what it would mean for something to be a philosopher 'at' a given Lewis-world, we find nothing that would allow any Lewis-world to play that role. According to one of these suggestions, something is a philosopher at a Lewis-world iff it has a counterpart that is part of that Lewis-world and is a philosopher; according to another, something is a philosopher at a Lewis-world iff it has a counterpart that is part of that Lewis-world, and all of its counterparts in that Lewis-world are philosophers. On neither suggestion is it plausible that there is any Lewis-world such that all and only the things that are philosophers (*simpliciter*) are philosophers 'at' it. In recognising questions about what things are like *simpliciter* which float free from questions about what things are like at Lewis-worlds, Lewis's view diverges very markedly from the standard modal B-theory. Indeed, it is not so clear that it is a version of the modal B-theory at all. Given that Lewis says nothing about any entities that play the full, standard "world" role, we must be very careful in attaching any significance to the fact that he is an egalitarian about the things he *calls* "possible worlds". Especially when we are working with technical jargon like 'possible world', there is a danger that insignificant verbal choices will create appearances of agreement that masks deep disagreement about the metaphysical questions we really care about, or appearances of disagreement that mask deep agreement. Chapters 3 and 4 will scrutinise Lewis's reasons for thinking that the concept of a Lewis-world has an important role to play in the proper development of a counterpart-theoretic account of modal claims, and conclude that it does not: modal counterpart theorists do better to stick with claims like MCT, in which nothing like the concept of a Lewis-world appears. This will put us in a better position to understand how modal counterpart theory relates to the dichotomy between modal A- and B-theories.

For analogous reasons, we will need to be very careful in thinking about how temporal counterpart theory relates to the dichotomy between temporal A- and B-theories. In the literature, temporal counterpart theorists generally go along with the standard identification of instants of time with certain three-dimensional slices through spacetime. Given the manifold picture (the preservation of which is a central motivation for temporal counterpart theory), this results in agreement with the temporal egalitarian's claim that 'all instants of time are on a par'. However, this verbal agreement is potentially very misleading, because of the disparities between the role that hypersurfaces play in temporal counterpart theory and the standard "instant" role

outlined in section 1.3. For example, the standard role allows us to say that for absolutely every x , x is a philosopher iff x is a philosopher at the present instant. But there is no plausible account which a temporal counterpart theorist could give of what it means for something to be the case ‘at’ a three-dimensional slice that would allow any single slice to play the role of “the present instant” in this formula. After we consider, in the next two chapters, how the basic idea of temporal counterpart theory can best be developed into a complete account of the temporal operators, we will be better placed to understand how the view relates to our dichotomy between temporal A- and B-theories.

The basic idea of modal counterpart theory has outrageous, astonishing consequences. It entails, for example, that if it is possible for me to be devoured by a fire-breathing dragon, then someone *is* devoured by a fire-breathing dragon.³⁵ Thus, one cannot coherently be very confident that modal counterpart theory is true, and very confident that it is possible for me to be devoured by a fire-breathing dragon, while being much less confident that there are any fire-breathing dragons. Lewis’s reaction was to retain his confidence in the relevant possibility claim, and to become much more confident that there are fire-breathing dragons (although probably not spatiotemporally related to us). Another possible reaction (Divers 2004) is to keep one’s degree of confidence that there are fire-breathing dragons fairly low, and commensurately reduce one’s confidence that it is possible for me to be devoured by one. Of course, we might acquire empirical evidence which, quite independently of counterpart theory, supports the hypothesis that some people are devoured by fire-breathing dragons. In that case, there will be somewhat less divergence between the credences required by modal counterpart theory and the credences that would seem reasonable to opponents of modal counterpart theory. Indeed, it is arguable that we already have such evidence. Certainly there are strong empirical physical reasons to take seriously accounts of the physical world which entail that the realm of ordinary objects is vastly more populous and more various than people normally think, and perhaps even that it includes person-devouring, fire-breathing dragons. Theories with this feature include various kinds of many-worlds interpretations of quantum mechanics, as well as the much

³⁵I am assuming that ‘devoured by a fire-breathing dragon’ counts as a qualitative predicate.

more readily intelligible hypothesis that the universe is infinite in spatiotemporal extent.³⁶ One might think that although modal counterpart theory is not very credible *a priori*, our empirical evidence in favour of such theories also provides strong *a posteriori* support to modal counterpart theory.³⁷ The ontologies of the relevant physical theories are not so *very* abundant as to provide instances of *every* qualitative property whose instantiation is metaphysically possible according to ordinary modal opinion—for example, they do not include bodies that travel faster than light. Thus, even given the evidence that supports such theories, many philosophers will still think that the rational packages of credences about modal and non-modal matters are packages which require one's credence in modal counterpart theory to be very low. But it might be argued that there is a relevant epistemological difference between our inclination to judge that it is metaphysically possible for someone to be devoured by a dragon and our inclination to judge that it is metaphysically possible for someone travelling through a vacuum to overtake a beam of light, and that we should be willing to revise the latter so long as we can hold on to the former.

No matter how abundant we take reality to be, the basic idea of modal counterpart theory still conflicts with some *modal* opinions that many of us would be loath to give up. It seems is possible that the following should be the case: there are no fire-breathing dragons (unrestrictedly speaking), and yet it is possible for me to be devoured by a fire-breathing dragon. While this is a relatively recondite claim (insofar as it involves iterated modal operators), it is nevertheless quite compelling on reflection. And it conflicts head-on with MCT, since it is obviously *not* possible for the following to be the case: there are no fire-breathing dragons (unrestrictedly speaking) and some counterpart of mine is devoured by a fire-breathing dragon. Thus, no matter how populous and various one might take reality to be, accepting modal counterpart theory requires some pretty astonishing, incredulous-stare-attracting, *modal* claims.

(I should note that some philosophers who do not accept the outrageously radical claim I am calling 'the basic idea of modal counterpart theory' still

³⁶In order to make it probable that there are dragons, the latter hypothesis has to be supplemented with the idea that probabilistic correlations between spatially separated regions drop off to zero as the spatial distance goes to infinity. Surprisingly few philosophers seem to realise that a spatially infinite universe is regarded as a live possibility in cosmology: this may arise from a misunderstanding of the Big Bang as a point-like event.

³⁷**Cite Wilson DPhil thesis**.

think of themselves as 'counterpart theorists'. They use 'counterpart theory' to stand for some more abstract theoretical form which is exemplified by MCT, but can also be exemplified by a more ontologically conservative theory on which 'there are dragons' does not follow from 'I could be devoured by a dragon'. Section *** below will briefly investigate what the ontologically conservative form of counterpart theory might amount to. But my main focus is on full-blooded, ontologically radical modal counterpart theories like Lewis's, that accept something in the vicinity of MCT. For I am mostly interested in modal counterpart theory because of the analogy with temporal counterpart theory, and in the temporal case, it is TCT that holds out the promise of avoiding both the argument from spacetime and the primordial objection to the B-theory.)

Just as acceptance of modal counterpart theory forces a certain kind of fit between our beliefs about how things *could be* and our beliefs about how things are, acceptance of temporal counterpart theory forces a certain kind of fit between our beliefs about how things *have been* and *will be* and our beliefs about how things are. For example, if you endorse TCT and you think that North America used to be a continent populated by dinosaurs, you must think that *some* continent *is* populated by dinosaurs (assuming that 'continent populated by dinosaurs' is a qualitative predicate). If you think that Obama used to be a community organiser and future president, you must think that there is someone who *is* a community organiser and future president. If you think that for every real number *n* less than your current age in years, you have been *n* years old, then you must think that for each such *n*, there is some person who *is n* years old, which means that the total number of people must be uncountably infinite. These are strong claims, which many will find utterly astonishing. However, temporal counterpart theorists will argue that the ontological claims in question can be justified by the combination of an empirically well-supported spacetime metaphysics with philosophically well-supported accounts of *what it is* to be a continent, to be populated by dinosaurs, to be a future president, to be *n* years old, etc. If those accounts are defensible, temporal counterpart theorists will be better equipped than modal counterpart theorists in responding to the objection that we lack empirical reason to accept the large ontology of ordinary objects that the theories require when combined with ordinary temporal or modal beliefs.

There will still remain a *modal* objection to temporal counterpart theory, whose premise is just that it is *possible* for there to be a continent that was

once populated by dinosaurs without there being any continent that *is* populated by dinosaurs. The existence of a vast multiplicity of continents is not, intuitively, a necessary condition for the existence of a continent that has had and will have a vast multiplicity of qualitative properties. But this modal objection seems weaker than the parallel objection to modal counterpart theory. One might think that the empirical evidence in favour of the manifold picture is evidence that all sorts of familiar facts *supervene* on the fundamental facts about the distribution of physical fields in the manifold, even though the acceptance of such supervenience requires rejecting certain seemingly coherent scenarios as impossible. It is quite natural to think that the discovery of spacetime reveals something important about the *nature* of time, and thus of temporal facts like the fact that North America was once populated by dinosaurs. By contrast, in the modal case, supervenience is trivial (assuming S5, all modal truths are necessarily true and hence supervene on everything); there is much less temptation to think of the nature of modal facts as a mystery that we might shed light on by empirical methods.

So: although modal and temporal counterpart theory both have some highly counterintuitive consequences, it will be worth while suspending disbelief for a while in order to investigate how these theories should best be developed. This will be the task of the next two chapters.

Chapter 3

Counterpart theory and the fixity of the qualitative

As you are now, so once was I
As I am now, so will you be

Tomb inscription, attributed to
Horace

3.1 Basic counterpart theory

According to what chapter 2 called the “basic idea” of modal counterpart theory, *to be possibly a certain qualitative way is to have a counterpart which is that way*. For example: to be possibly staring at a blue swan is to have a counterpart who is staring at a blue swan. To avoid the difficult issues raised by the quantification over ‘ways’, it will be helpful to consider a version of the proposal in the form of a schema:

Basic Modal Counterpart Theory (BMCT:)

$$\Diamond\phi(\alpha) := \exists x(Cx\alpha \wedge \phi(x))$$

$$\Box\phi(\alpha) := \forall x(Cx\alpha \rightarrow \phi(x))$$

Here, ‘ $\phi(\alpha)$ ’ is to be replaced with a sentence in which all vocabulary is qualitative, except for one or more occurrences of a singular term α (which could be a name or a free variable). ‘ $\phi(x)$ ’ stands for the result of replacing all the occurrences of α in $\phi(\alpha)$ with some variable x which does not already occur in $\phi(\alpha)$. ‘ $Cx\alpha$ ’ abbreviates ‘ x is a counterpart of α ’. Remember that this is functioning as a placeholder, which will be cashed out in different ways corresponding to different resolutions of the context-sensitivity of ‘possibly’ and ‘necessarily’. Finally, ‘ $:=$ ’ indicates the giving of an analysis: perhaps it

may be rendered in English as ‘for it to be the case that ... is for it to be the case that ...’.

Similarly, the “basic idea” of temporal counterpart theory is that to be *going to be* a certain qualitative way is to have a future-counterpart which is that way, and to *have been* a certain qualitative way is to have a past-counterpart which is that way. Again, it will be helpful to consider this proposal in schematic form:

Basic Temporal Counterpart Theory (BTCT):

$$\mathbf{F}\phi(\alpha) := \exists x(\mathbf{C}_F x\alpha \wedge \phi(x))$$

$$\mathbf{G}\phi(\alpha) := \forall x(\mathbf{C}_F x\alpha \rightarrow \phi(x))$$

$$\mathbf{P}\phi(\alpha) := \exists x(\mathbf{C}_P x\alpha \wedge \phi(x))$$

$$\mathbf{H}\phi(\alpha) := \forall x(\mathbf{C}_P x\alpha \rightarrow \phi(x))$$

The predicates ‘ $\mathbf{C}_F$ ’ (‘future-counterpart’) and ‘ $\mathbf{C}_P$ ’ (‘past-counterpart’) here are placeholders, which temporal counterpart theorists are free to analyse in different ways.

(One plausible constraint on such analyses is worth mentioning, however, namely that ‘ $\mathbf{C}_F xy$ ’ and ‘ $\mathbf{C}_P yx$ ’ should turn out to be equivalent: a thing’s past-counterparts are the things whose future-counterpart it is. Imposing this constraint is a natural way to validate (for the cases to which BTCT applies) the basic tense-logical “mixing” axioms, $\phi \rightarrow \mathbf{HF}\phi$ —‘What is the case has always been going to be the case’—and $\phi \rightarrow \mathbf{GP}\phi$ —‘what is the case always will have been the case’.)

The requirement that α be the only referential item in $\phi(\alpha)$ must be interpreted so as to exclude, for example, ‘Merkel is German’ as a legitimate substitution instance for ‘ $\phi(\alpha)$ ’. Modal counterpart theorists should want to be able to say that it is possible that Merkel is German and has a collection of blue swans while denying that anyone *is* German and has a collection of blue swans. The problem with the predicate ‘German’ is that it is non-qualitative: it is (roughly) equivalent to ‘from Germany’. For BMCT to be defensible, it must be restricted to a sublanguage in which the job of *referring* is done only by singular terms, and everything else is qualitative. And BTCT, too, is much more defensible when it is understood as subject to this restriction. To get an analysis of ‘It is possible that Merkel is German and has a collection of blue swans’ or ‘It will be the case that Merkel is German and not a chancellor’, we will first have to analyse away ‘is German’ in favour of something like ‘from Germany’, and then apply some (as yet unseen) generalisation of BMCT or

BTCT which can deal with cases in which multiple referential items appear within the scope of modal operators.¹

BMCT and BTCT obviously are not complete accounts of the meaning of modal and temporal operators, since they do not tell us how to make sense of the application of modal or temporal operators to sentences that contain no referential items at all, or more than one. The latter case, in which multiple referential items occur in the scope of a single modal or temporal operator, raises tricky issues best postponed until chapter 4. The former case, where a modal or temporal operator applies to a sentence built up entirely out of qualitative vocabulary, with no referential items at all, is easier to deal with.

For these sentences, as we will shortly see, acceptance of BMCT practically forces us to accept the following highly counterintuitive claim:

Modal Redundancy: When ϕ is a closed sentence built up entirely out of qualitative vocabulary, $\Box\phi$ and $\Diamond\phi$ are equivalent to ϕ .

In other words, there is no contingency at all as regards the purely qualitative facts. If it is contingently the case that ϕ , then the fact that ϕ must be a fact about at least one specific object. In slogan form: *All contingency is de re contingency*. Likewise, acceptance of BTCT practically forces us to accept

Temporal Redundancy: When ϕ is a closed sentence built up entirely out of qualitative vocabulary, $F\phi$, $P\phi$, $G\phi$ and $H\phi$ are all equivalent to ϕ .

¹Special difficulties arise when the term α in $\phi(\alpha)$ occurs in the scope of a propositional attitude verb. If we take the name 'Hesperus' to be a referential term and hence an admissible substituent for ' α ', BMCT entails the equivalence of (i) and (ii):

- (i) Necessarily $\forall x\forall y(y = \text{Hesperus} \rightarrow (x \text{ believes that Hesperus is bright} \leftrightarrow x \text{ believes that } y \text{ is bright}))$
- (ii) $\forall z(z \text{ is a counterpart of Hesperus} \rightarrow \forall x\forall y(y = z \rightarrow (x \text{ believes that } z \text{ is bright} \leftrightarrow x \text{ believes that } y \text{ is bright})))$

(ii) is an uncontroversial logical truth, but (i) is quite controversial: many theories about the behaviour of names in attitude ascriptions deny that they are "transparent" in this way (For discussion, and a defence of (i), see Dorr 2014). I count proponents of these theories as denying that 'Hesperus' is a referential singular term, in the sense relevant to what counts as a substitution instance of BMCT. Most of these proponents will *also* deny that 'Hesperus' is purely qualitative; in that case, they will not be able to use BMCT as a general analysis of modal claims involving 'Hesperus'. Nevertheless, they might endorse it as an analysis of modal claims in which 'Hesperus' occurs only outside the scope of attitude verbs, while leaving the task of analysing modal sentences that embed attitude ascriptions for another occasion.

If it is temporarily the case that ϕ , the fact that ϕ must be a fact about some particular object or objects. *All change is de re change.*

Let us see how BMCT and BTCT push us towards these surprising redundancy claims. Assume that there is at least one object that has a counterpart (perhaps itself). Let ' a ' name some object that has a counterpart (e.g. Mars), and let ' F ' be some qualitative predicate (e.g. 'red'). While BMCT does not tell us how to analyse $\Diamond\phi$ when ϕ is composed of purely qualitative vocabulary, it does yield something nearly as good, namely:

$$\Diamond(\phi \wedge (Fa \vee \neg Fa)) := \exists x(Cxa \wedge (\phi \wedge (Fx \vee \neg Fx)))$$

The right hand side of this analysis logically entails ϕ , and indeed it is equivalent to ϕ given our assumption that a has a counterpart. Thus, if we wanted to deny that $\Diamond\phi$ is equivalent to ϕ while accepting BMCT, we would have to deny that $\Diamond\phi$ is equivalent to $\Diamond(\phi \wedge (Fa \vee \neg Fa))$. But this is very implausible, since $\phi \wedge (Fa \vee \neg Fa)$ is tautologically equivalent to ϕ , and tautologically equivalent sentences can be substituted *salva veritate* in the scope of modal operators. So it looks like acceptance of BMCT commits us to regarding ϕ as equivalent to $\Diamond\phi$ whenever ϕ is purely qualitative. (Likewise, *mutatis mutandis*, for BTCT and temporal operators.)

One could resist this reasoning by rejecting the equivalence of $\Diamond\phi$ and $\Diamond(\phi \wedge (Fa \vee \neg Fa))$. While this equivalence is extremely plausible, and valid according to orthodox modal logic, there are systems of modal logic (e.g. system Q from Prior 1957) in which it fails, and indeed counterpart theory has often been linked with a rejection of such equivalences. This would make room for a view that combines BMCT with contingency in the qualitative facts, or BTCT with temporariness in the qualitative facts. But if the aim was not merely to reject *Modal Redundancy* or *Temporal Redundancy*, but to do justice to our intuitive beliefs about the extent of contingency and change, this approach is deeply inadequate. Typically, when we are pre-theoretically inclined to think that it is contingently or temporarily the case that ϕ (where ϕ is purely qualitative), we are just as strongly inclined to accept various claims to the effect that it could be or will be the case that (not- ϕ and $\psi(\alpha)$), for some referential term α . For instance, we do not merely think that it is possible that no white cubes have sides one metre long (a qualitative claim); we think that it is possible that (no white cubes have sides one metre long, and Mars is red). BMCT directly rules out the latter claim: if some white cubes have sides one metre long, then of course Mars has no red

counterparts that are such that no white cubes have sides one metre long. Thus, for those who are committed to the intuitive judgments which conflict with *Modal Redundancy* or *Temporal Redundancy*, accepting BMCT or BTCT is not a viable option.

The idea that there is no contingency, or no change, as regards purely qualitative matters looks *prima facie* bizarre. Given the discovery that a literal-minded interpretation of the “basic ideas” of modal and temporal counterpart theory (BMCT and BTCT) commit us to these strange claims, the natural reaction is to look for some way of modifying these formulations, so as to allow them to fit with some view on which modal/temporal operators need not be vacuous when applied to purely qualitative sentences. Inspiration for this project can be drawn from the versions of modal counterpart theory developed by Lewis, in Lewis 1968 and Lewis 1986b. For even when we restrict our attention to the case of singly *de re* modal claims, Lewis’s analysis differs from BMCT in a way that looks extremely relevant: unlike BMCT, Lewis’s analysis involves talk about the entities which he calls “possible worlds”, and which I am calling “Lewis-worlds”: in the final analysis (Lewis 1986b, pp. 75–6), objects any two parts of which stand in some spatiotemporal, or “analogically spatiotemporal”, relation, and which are not parts of any other such objects.

In the case of singly *de re* claims, which is our present concern, this works out as follows:

Restrictive Counterpart Theory (RCT)

$$\Diamond\phi(\alpha) := \exists w\exists x(Ww \wedge Ixw \wedge Cx\alpha \wedge [\phi(x)]^w)$$

$$\Box\phi(\alpha) := \forall w\forall x(Ww \wedge Ixw \wedge Cx\alpha) \rightarrow [\phi(x)]^w$$

Here, w and x are some new variables not already in $\phi(\alpha)$. ‘ Ww ’ means ‘ w is an accessible Lewis-world’: ‘accessible’ is a placeholder which may be filled in in different ways, depending on how we are resolving the context-sensitivity in ‘possibly’ and ‘necessarily’. ‘ Ixw ’ means ‘ x is in w ’. $\phi(x)$ is, as before, the sentence derived from $\phi(\alpha)$ by replacing each occurrence of α with one of x . $[\phi(x)]^w$ is derived from $\phi(x)$ by adding a restricting clause ‘in w ’ to every quantifier in $\phi(x)$ that is not in the scope of any other modal operator. So, for example, if $\phi(\alpha)$ is ‘Some people are taller than α ’, $[\phi(x)]^w$ is ‘Some people in w are taller than x ’; if $\phi(\alpha)$ is ‘Most geese are the same colour as some animal that belongs to α ’, $[\phi(x)]^w$ is ‘Most geese in w are the

same colour as some animal in w that belongs to x' . (A recursive definition implementing this operation for a certain formal language is given in 1968, but for our present purposes this informal presentation will do.)²

There is a natural way to generalise RCT to claims in which modal operators apply to purely qualitative sentences: what Lewis calls '*de dicto*' modal claims. And because of the role played by quantification over Lewis-worlds and the quantifier-restriction operation in RCT, this generalisation does *not* have the result that such modal operators are vacuous. For claims of this form, Lewis's analysis can be given as follows:

$$\begin{aligned} \textit{Restriction} \quad \Diamond\phi &:= \exists w(Ww \wedge [\phi]^w) \\ \Box\phi &:= \forall w(Ww \rightarrow [\phi]^w) \end{aligned}$$

Thus, for example, *Restriction* tells us that for it to be possible that some swans are blue is for there to be an accessible Lewis-world such that some swans in it are blue; for it to be possible that most swans are the same colour as some goose is for there to be an accessible Lewis-world such that most swans in it are the same colour as some goose in it; and so on.

Just like BMCT, RCT and *Restriction* must be understood as restricted to a language in which singular terms are the only carriers of reference to particular items, thus excluding non-qualitative predicates like 'is German'. However, because of the quantifier-restricting operation, Lewis needs to impose an additional requirement on the predicates that can occur in legitimate instances of RCT and *Restriction*. The point of this requirement is to rule out 'implicitly quantificational' expressions like 'invent' and 'discover' (1986b, p. 6)—expressions which under analysis reveal further quantifiers for which the restricting operation could make a difference. Before we can apply RCT or *Restriction* to claims involving predicates of this sort, we must first replace them with analyses which make the relevant quantifiers explicit, so that we can add the relevant modifiers. Thus, 'Possibly, x invents bifocals' must first be replaced with 'Possibly, x thinks of bifocals and no-one thinks of bifocals before x' ', before we can apply RCT to get 'There is a Lewis-world w , and a counterpart y of x who is in w , such that y thinks of bifocals, and no-one in w thinks of bifocals before y .'

²I have left out an important qualification: according to Lewis (1986, p. 6), it is an optional matter, to be decided on pragmatic grounds, whether a restrictor should be attached to any given quantifier in the analysis of a modal claim. In section 3.4 I will consider how this qualification affects the plausibility of Lewis's analyses of modal sentences.

Let us call a predicate ‘well-behaved’ iff it is qualitative and *not* in the relevant way ‘implicitly quantificational’, so that it can be used in legitimate instances of RCT and *Restriction*. If one takes the idea of ‘implicit quantification’ very literally, it might seem that very few predicates are well-behaved: for example, we will not be able to use ‘triangle’ in instances of these schemas, since it has an analysis involving quantification (‘figure with three angles’). But there is no need to be as demanding as that. For a wide range of predicates which are necessarily equivalent to complex formulae involving quantification, the result of substituting those complex formulae and then inserting all the relevant restrictors will make no truth-conditional difference. Given the transitivity of parthood, this will be true of any predicate that expresses an *intrinsic* property or relation. When such predicates have analyses involving quantifiers, the quantifiers in question can always be taken to be restricted to the *parts* of a given object. And since parthood is transitive, the parts of an object in a certain Lewis-world are always themselves in that Lewis-world. So, for example, the claim that there is a Lewis-world *w* that contains a white triangle—i.e. a white plane figure *x* such that exactly three angles are parts of *x*—is equivalent to the claim that there is a Lewis-world *w* that contains a white plane figure *x* such that exactly three angles *in w* are parts of *x*. Thus, we can harmlessly take the class well-behaved predicates to include all predicates expressing intrinsic (and qualitative) properties and relations. Indeed, we could even construe ‘well-behaved’ so as to apply to certain predicates expressing *extrinsic* properties and relations, so long as their analyses only involve quantification over items related to the given item by some spatiotemporal, or analogically spatiotemporal, relation. For most purposes, we will not need to worry too much about how to take the restriction to well-behaved predicates, since Lewis gives us plenty of examples of predicates (like ‘blue swan’ and ‘talking donkey’) which he is willing to use in instantiating RCT and *Restriction*.

There are some simple examples in which the analyses yielded by RCT will plausibly turn out to be equivalent to those yielded by BMCT. Consider for example the sentence ‘It is possible that Obama be staring at a blue swan’. BMCT analyses this as ‘A counterpart of Obama is staring at a blue swan’, while RCT analyses it as ‘There is a Lewis-world such that a counterpart of Obama in it is staring at a blue swan in it’. It is plausible (at least given background mereological principles accepted by Lewis) that all blue swans are parts of Lewis-worlds, and that any two things one of which is staring at the other are part of the same Lewis-world. Modulo these assumptions,

the two analyses are logically equivalent. But there is certainly no guarantee that the quantifier-restrictions introduced by RCT will always end up being redundant in this way. For example, 'It is possible that Obama be staring at every blue swan' is analysed by BMCT as 'A counterpart of Obama is staring at every blue swan', whereas RCT analyses it as 'There is a Lewis-world such that a counterpart of Obama in it is staring at every blue swan in it'.

Although the machinery of quantification over Lewis-worlds and restriction of quantifiers to those worlds makes Lewis's analysis RCT more complicated than BMCT, the added complexity may initially seem to be a price worth paying in order to avoid the highly counterintuitive claim thesis of *Modal Redundancy*, which as we have seen goes hand in hand with BMCT. However, as we will see in the next section, the machinery also leads to some puzzles, which we will need to investigate before reaching a final judgment about which of the two accounts is superior.

3.2 Some puzzles for Lewis

This section will raise a trio of puzzles about Lewis's programme for analysing modal claims. These puzzles arise already for Lewis's analysis of '*de dicto*' modal claims (*Restriction*), although similar puzzles could be raised for his analysis of singly *de re* claims (RCT). In fact, the first and second puzzles will depend only on the following very elementary (English translations of) instances of *Restriction*, which Lewis uses as examples on the first few pages of *OPW*:

- (1) It is possible that some swans are blue := there is a Lewis-world such that some swans in it are blue.
- (2) It is possible that no swans are blue := there is a Lewis-world such that no swans in it are blue.

Here the restriction to 'accessible' Lewis-worlds is omitted, as Lewis thinks it should be when we are concerned with metaphysical possibility.

I have been using the symbol ' := ' to indicate the giving of an analysis. If we have any grip at all on what it means to give an analysis of something, we know that $\ulcorner \phi \text{:=} \psi \urcorner$ entails $\ulcorner \text{Necessarily, } \phi \text{ iff } \psi \urcorner$. If this is given up, it becomes mysterious how the meaning of (1) and (2) go beyond the corresponding material biconditionals.³ So for example, (1) entails (1')

³I remain neutral on the question whether $\ulcorner \phi \text{:=} \psi \urcorner$ is *equivalent* to $\ulcorner \text{It is metaphysically necessary that } (\phi \leftrightarrow \psi) \urcorner$. But I note that there is a reason for those, like Lewis, who seek a

- (1') Necessarily: it is possible that some swans are blue iff there is a Lewis-world such that some swans in it are blue.

Puzzle 1 depends on (1') alone. It seems obvious that it *is*, in fact, possible for it to be the case that no swans are blue. And it seems only a little less obvious that it is possible for it to be *contingently* the case that no swans are blue—i.e. that it is possible that (no swans are blue and it is possible that some swans are blue). But given (1'), this claim must be rejected. For in combination with (1') it entails that it is possible that (no swans are blue and there is a Lewis-world such that some swans in it are blue).⁴ And this cannot be possible, since it is contradictory: any blue swan that is part of a Lewis-world is a blue swan. This is surprising. Given that Lewis thought that there could have been no blue swans, could he really have been committed to denying that there could have *contingently* been no blue swans?

Puzzle 2 depends on the combination of (1') with (2). As we saw in *Puzzle 1*, (1') entails that if it is possible that no swans are blue, it is possible that it is not possible that some swans are blue. Suppose that—as Lewis believed—some Lewis-worlds don't contain blue swans, and some do. Then by (2), it is possible that no swans are blue. So by (1'), it is possible that it is not possible that some swans are blue. But it is also possible that some swans are blue, by (1). This combination of claims is inconsistent in the modal logic S5, where what is possibly necessary is simply necessary, and what is possibly impossible is simply impossible. It is thus puzzling to find it seeming to follow so directly from Lewis's commitments. For Lewis generally seems to be friendly towards S5, at least when the modal operators are interpreted as expressing metaphysical necessity and possibility and the sentences we are dealing with are "*de dicto*" (purely qualitative), as 'there are blue swans' and 'there are no blue swans' are supposed to be.

Puzzles 1 and 2 turn on the assumption that the correctness of an analysis requires the necessarily equivalence of the analysans and the analysandum. *Puzzle 3* does not require this assumption. It turns on another instance of *Restriction*:

- (3) It is possible that there are two things that are not analogically spatiotemporally related := there is a Lewis-world such that there are two things in it that are not analogically spatiotemporally related.

general analysis of modality to answer yes: it wouldn't be so exciting if one could analyse modality, counterfactuals, etc. in non-modal terms, but still had to take ':= ' as a primitive, manifestly non-extensional, operator.

⁴The argument here is of the evidently valid form $\Box(\phi \leftrightarrow \psi), \Diamond(\xi \wedge \phi) \vdash \Diamond(\xi \wedge \psi)$.

The right-hand side of (3) is false, since by definition, a Lewis-world is a maximal object any two parts of which are analogically spatiotemporally related. Thus, anything that has two parts which are not analogically spatiotemporally related is, ipso facto, not a Lewis-world. So if we accept (3), we have to say that it is *not* possible for there to be two things that are not analogically spatiotemporally related. This sits oddly (to say the least!) with Lewis's notorious claim that there *are* many Lewis-worlds, which of course entails that there are many things that are not analogically spatiotemporally related. Whatever *is* the case *could* be the case. This is a version of the modal axiom T, and plays an extremely basic role in our practice of reasoning with modal operators on many of their interpretations, including the metaphysical interpretation with which we are presently concerned. While no principle in logic is so uncontroversial as to be beyond all debate, the idea that some things that are necessarily the case are not the case, or that some things that are the case could not be the case, takes us deep into the alien realm of logical heresy.⁵

The obvious move for Lewis to make in response to these puzzles is to invoke the familiar idea that quantification is sometimes tacitly restricted. You can say 'There is no beer' without asserting anything false even if there is beer but none of it is in your house, or even if there is beer in your house but only in the form of a puddle of spilt beer on the floor of the fridge. Similarly, you can say 'There are no blue swans' and assert nothing false even if (as Lewis believed) there are blue swans but none of them is in the Lewis-world of which you are part.

(We need not concern ourselves with the question how the phenomenon of quantifier domain restriction fits into compositional semantics. I will write

⁵Hudson (1997) and Parsons (2009) also note the apparent failure of T induced by Lewis's analysis. Jeff Russell pointed out to me that if we interpret Lewis as committed to the rejection of T, we can still give him the related schema $\Box\phi \rightarrow \text{ACT}\phi$: what is necessarily the case is *actually* the case. Lewis could thus diagnose our temptation to think that what is necessarily the case is the case as the result of our mistaken assumption that what is actually the case is the case. This is probably the right way for the rejecter of T to go, and there is certainly textual evidence for attributing certain claims of the form ' ϕ but actually not- ϕ ' to Lewis. But the package in which neither necessity nor actuality is factive is still very alien. When people utter (3), are they right but actually wrong? We would normally take it that if David spoke truly, then David would have got a prize if everyone who actually spoke truly had got a prize; but we will not be able to say this if we think that uttering (3) involves speaking truly without actually speaking truly.

as if it involves a semantic context-sensitivity in determiners like ‘some’; but everything I say should be acceptable, *mutatis mutandis*, to those (e.g. Stanley and Szabó 2000) who locate the context-sensitivity in nouns, or in some silent syntactic elements associated with nouns.⁶)

The natural way to resolve the puzzles is to say that the analyses that generate them involve a mixture of restricted and unrestricted quantifiers. Consider (1) and (2), for example:

- (1) It is possible that some swans are blue := there is a Lewis-world such that some swans in it are blue.
- (2) It is possible that no swans are blue := there is a Lewis world such that no swans in it are blue.

The proposal is that the true readings of these claims are readings in which the quantifiers on the left (‘some’ and ‘no’) are somehow tacitly restricted, whereas the quantifiers on the right (‘there is’, ‘some’ and ‘no’) are unrestricted. We can make it clear that these are the intended readings by adding subscripts to our quantifiers: ‘u’ to indicate unrestricted quantification, and ‘r’ to indicate the relevant kind of restriction, whatever it turns out to be. Then the proposed interpretations of (1) and (2) will be represented as follows:

- (1_r) It is possible that some_r swans are blue := there is_u a Lewis world such that some_u swans in it are blue.
- (2_r) It is possible that no_r swans are blue := there is_u a Lewis world such that no_u swans in it are blue.

Of course, the proposal is not restricted to these particular analyses, but extends to all instances of *Restriction*:

$$\begin{aligned}\Diamond\phi &:= \exists w(Ww \wedge [\phi]^w) \\ \Box\phi &:= \forall w(Ww \rightarrow [\phi]^w)\end{aligned}$$

The idea is that these sentences are true only when all quantifiers on the left are all tacitly restricted in some distinctive way, while the quantifiers on the right are unrestricted. The same interpretation could be applied to instances of RCT.

⁶Even those like Bach (1994) who regard the phenomenon as pragmatic rather than semantic should be able to reconstruct my discussion to fit with their view.

This way of understanding *Restriction* makes the puzzles go away.

Puzzle 1: (1_r) does not make 'possibly some_r swans are blue' inconsistent with 'no_r swans are blue', only with 'no_u swans are blue'. Thus it is consistent with the truth of 'Possibly, it is contingently the case that there are no_r blue swans'.

Puzzle 2: for the same reason, (2_r) does not license the inference from 'Possibly, no_r swans are blue' to 'Possibly, it is not possible that some_r swans are blue'. So the inference to the S5-inconsistent 'Possibly there are no blue swans, and possibly, it is not possible for there to be blue swans' (on either interpretation) is blocked.

Puzzle 3: The intended interpretation of (3), with restricted quantifiers on the left, tells us that it is not possible that there be_r two analogically spatiotemporally unrelated objects. It follows by T that any_r two objects are analogically spatiotemporally related. This tells us something interesting about the character of the relevant form of restriction represented by 'r'. But it is perfectly consistent with Lewis's characteristic claim that there are many_u Lewis-worlds, and hence many_u analogically spatiotemporally unrelated things.

So, provided we understand *Restriction* as a schema for analysing possibility claims involving some variety of restricted quantification, trouble is averted.

However, this response raises a further question. Given that 'some' in 'Some swans are blue' admits of both unrestricted and restricted readings, it is extremely plausible that 'some' in 'It is possible that some swans are blue' also admits of both kinds of readings. So, how should we analyse 'It is possible that some swans are blue' on the unrestricted reading of its quantifier? More generally, how should we analyse claims in which unrestricted quantifiers occur in the scope of modal operators? If this is a good question, any general programme for analysing modal claims in non-modal terms had better be able to answer it.

It is hard to see how it could fail to be a good question. In general, while embedding a context-sensitive sentence inside some larger sentence often makes a difference as regards which of its readings we find most natural and salient, readings that are available when a sentence is asserted on its own do not just disappear under embedding. A semantic mechanism that made it flat-out impossible to interpret any quantifier embedded under a modal operator as unrestricted would be strange and unprecedented.

The setup in Lewis 1968 might suggest such a prohibition. There, the project of reducing modality is cast as one of finding translations from one formal language, which contains modal operators, into another, which lacks them. The output language is not just the modal-operator-free fragment of the input language, since even sentences with no modal operators are translated non-homophonically. For example, ' $\exists xFx$ ' in the input language is translated as ' $\exists x(Ix@ \wedge Fx)$ ' in the output language (' $Ix@$ ' means ' x is in the actual world'). This non-homophonic treatment of ' $\exists x$ ' suggests that the quantifier-meanings in the input language are restrictions of the quantifier-meanings in the output language. If one thought that the input language had to be capable of representing every admissible interpretation of English sentences involving a modal operator, one would be pushed to the view that quantifiers in English cannot be interpreted as unrestricted when they occur in the scope of a modal operator.

But such a prohibition is quite unmotivated. If unrestricted quantification were a rare practice engaged in only by philosophical initiates, perhaps there would be some temptation to think that this special way of talking cannot meaningfully be combined with ordinary modal operators. But in fact, ordinary people hold, and express, lots of opinions about what there is, unrestrictedly speaking. For example, they are very confident that no_u bachelors are married, and fairly confident that no_u swans are blue. Lewis, of course, thought that many ordinary negative judgments involving unrestricted quantification were *mistaken*. But if ordinary people can express unrestricted quantification when they say things like 'No swans are blue', surely they can also express unrestricted quantification when they say things like 'It is possible that no swans are white' and 'It is necessary that no bachelors are married'. Ordinary usage gives no support at all to the claim that such readings are semantically prohibited.

Moreover, it is hard to see how any prohibition of this sort could work. A restricted quantifier Q_r is presumably equivalent to the combination of an unrestricted quantifier Q_u with a predicate R —an invented one, if necessary—expressing the relevant restriction.⁷ For example, ' $\text{Most}_r Fs \text{ are } Gs$ ' will be equivalent to ' $\text{Most}_u Fs \text{ which are } R \text{ are } Gs$ '. But if ' $Q_r Fs \text{ are } G$ ' is equivalent to ' $Q_u Fs \text{ which are } R \text{ are } G$ ', 'It is possible that $Q_u Fs \text{ which are } R \text{ are } G$ ' surely should make sense, and be equivalent to 'It is possible that $Q_r Fs \text{ are } G$ '. We don't want to dismiss 'It is possible that $Q_u Fs \text{ which are } R \text{ are } G$ ' as nonsense just on the grounds that it contains an unrestricted quantifier in the scope of

⁷This certainly was Lewis's view of restricted quantification.

a modal operator. If so, the mechanism that enforces the prohibition must somehow check to make sure that every unrestricted quantifier in the scope of a modal operator ends up combining with a restrictor of the right kind—a complicated and messy task. Worse, there are sentences—for example, ‘Possibly some swans are blue’ on the restricted reading of ‘some’—whose analyses *require* unrestricted quantification. If we want ‘Possibly, possibly, some swans are blue’ to be legitimate and ‘Possibly, there is a Lewis-world such that some swans in it are blue’ to be prohibited on the unrestricted reading of ‘some’, the mechanism that enforces the prohibition will have to distinguish, using some obscure mechanism, between analysis and analysandum. Even if there were such a mechanism, it would be an arbitrary imposition, which we could easily stipulatively remove for philosophical purposes, if we found it helpful to do so.

So a general program for analysing modality *does* need an answer to the question how to analyse claims in which unrestricted quantifiers occur in the scope of modal operators, such as ‘Possibly, some_u swans are blue’. And for someone who accepts *Restriction* as an analysis of modal claims involving restricted quantification, the possible answers to this question are severely limited. In fact, for a certain class of claims, namely those derived from well-behaved predicates by applying an unrestricted *existential* quantifier, there is essentially only one possible answer:

Existential Redundancy: When F is well-behaved, $\Box\exists_u xFx$ and $\Diamond\exists_u xFx$ are both equivalent to $\exists_u xFx$.

The claim that $\exists_u x\phi(x)$ entails $\Diamond\exists_u x\phi(x)$, and that $\Box\exists_u x\phi(x)$ entails $\exists_u x\phi(x)$, should be uncontroversial, since they are consequences of T. As for the converse entailments, suppose that (like almost all philosophers other than Lewis) one took the claim that nothing_u is F to be consistent with the claim that possibly, something_u is F . Then what on earth could motivate one to accept the necessity of (4), as required by our current way of understanding *Restriction*?

$$(4) \quad \Diamond\exists_r xFx \rightarrow \exists_u xFx$$

For any kind of restriction we antecedently understand, (4) is profoundly counterintuitive. If the proposal is to be worth bothering with at all, it must be because it is an inevitable upshot of some general strategy for analysing modal claims in non-modal terms. But if we were in possession of some such strategy which allowed us to avoid the counterintuitive consequences

of *Existential Redundancy*, it would simply be perverse not to use the same trick to avoid the (equally counterintuitive) consequences of (4).

Given *Existential Redundancy*, which says that certain modal operators are redundant when applied to sentences built up in certain ways from well-behaved predicates and unrestricted quantifiers, it is natural to consider the following stronger thesis:

Generalized Redundancy: Whenever ϕ is a closed sentence built up from well-behaved predicates, unrestricted quantifiers, and truth-functional connectives, 'Possibly ϕ ' and 'Necessarily ϕ ' are both equivalent to ϕ .

One can certainly imagine views which accept *Existential Redundancy* while resisting *Generalized Redundancy*, allowing (for example) that even if some_u swans are blue, it is still possible that no_u swans should be blue. One way to formulate a view of this sort to keep the letter of *Restriction*, understood once again as an analysis of modal claims involving unrestricted quantification, while relaxing our previous stipulations about the meaning of its primitive predicate ' Ww ' (which we formerly understood to mean ' w is an accessible Lewis-world'):

$$\begin{aligned}\Diamond\phi &:= \exists w(Ww \wedge [\phi]^w) \\ \Box\phi &:= \forall w(Ww \rightarrow [\phi]^w)\end{aligned}$$

Suppose we just read ' Ww ' as ' w is an appropriate part of reality', and consider what we get when we plug in different accounts of appropriateness. One thing we can immediately see is that if we want T to be valid, the whole of reality—something such that everything whatsoever counts as being 'in' it—had better count as 'appropriate'. If it does, and if we treat the existence of such an all-encompassing entity as a logical truth, our analysis will preserve the entailments from $\Box\phi$ to ϕ , and from ϕ to $\Diamond\phi$. If not, not.⁸ Note that so long as we do hold fixed that the whole of reality is an appropriate part of reality, we will recover *Existential Redundancy*: $\Diamond\exists xFx$ is equivalent to $\exists w(Ww \wedge \exists x(Ixw \wedge Fx))$, which is equivalent to $\exists xFx$ so long as $\exists w(Ww \wedge \forall xIxw)$ can be treated as a logical truth. What else should we count as an 'appropriate part of reality'? One possible view is that the only

⁸In principle, one could preserve T without counting the whole of reality as 'appropriate', so long as one's account of appropriateness guaranteed that some part of reality which is an intrinsic duplicate of the whole of reality is appropriate. (Recall that objects that are infinite in size can have proper parts that are intrinsic duplicates of themselves.)

other appropriate parts, besides the whole of reality, are the Lewis-worlds.⁹ But this restriction is arbitrary. If there could be_u many Lewis worlds, and there could be_u exactly one, what motivation could there be for denying that there could be_u two, or any other intermediate number? It seems better to understand 'appropriate' in such a way that arbitrary fusions of appropriate parts of reality are themselves appropriate.¹⁰

However, this whole approach seems pointless. For if anything short of reality as a whole counts as "appropriate", we will no longer have an S5 modal logic, even for claims involving just unrestricted quantifiers applied to well-behaved predicates. For example, suppose that there is an appropriate part that lacks blue swans, so that it is possible that there are no_u blue swans. By *Existential Redundancy*, 'it is possible that there should be_u blue swans' is equivalent to 'there are_u blue swans'; so we are committed to the claim that it is possible that it is not possible that there should be_u blue swans. Reality could be "smaller" (it could lack blue swans), but it is necessary that it could not be bigger (necessarily, if it lacks blue swans, it couldn't contain them); once the blue swans are gone, they cannot come back. Endorsing such a claim for the sake of being able to agree with ordinary people when they claim that it is possible that no_u swans are blue seems like a bad trade. For once one accepts *Existential Redundancy*, one is already committed to rejecting a vast array of common-sense modal claims involving unrestricted quantification, for example the claim that it is possible that it is contingently the case that no_u swans are blue. It seems perverse to jettison the compelling logical principles that constitute S5 merely in order to slightly decrease the scope of this systematic disagreement as regards particular modal claims.

The 'appropriate parts of reality' approach was only one way of combining *Existential Redundancy* with the denial of *Generalized Redundancy*. Might there be some other, more attractive, way of developing such a combination? Well, I certainly can't think of one. What's more, there are some considerations which suggest that any such combination will have to be, in effect, a version of the 'appropriate parts' view. Let us assume that it is necessary that there is such a thing as "the whole of reality", something which absolutely

⁹This version of *Restriction* is equivalent to the "T-preserving analysis" discussed by Parsons (2009).

¹⁰Bricker (2001) endorses something that looks like *Restriction* with '*Ww*' interpreted to mean '*w* is a fusion of Lewis-worlds'. However, he does not intend this analysis to apply to possibility claims involving absolutely unrestricted quantifiers. I will discuss Bricker's account of the relevant kind of restriction in section 3.3 below.

everything is “in”: $\Box \exists_u x \forall_u y (Iyx)$. Then for any closed sentence ϕ built up from well-behaved predicates, unrestricted quantifiers, and truth-functional connectives, we have that $\Box(\phi \rightarrow \exists_u w[\phi]^w)$: if ϕ is true, there is bound to be at least one thing (viz. the whole of reality) such that ϕ remains true when its quantifiers are restricted to parts of that thing. So from $\Diamond\phi$, we can conclude that $\Diamond\exists_u w[\phi]^w$. If we could treat $[\phi]^w$ as the result of applying a *well-behaved* predicate to w , we could apply *Existential Redundancy* at this point, concluding that $\exists_u w[\phi]^w$. This would bring us back to the ‘appropriate parts’ account: the only way for ϕ to be possibly true is for there to be something such that ϕ is in fact true when its quantifiers are restricted to its parts. So, can we treat $[\phi]^x$ as equivalent to the result of applying a well-behaved predicate to x ? If we take the well-behaved predicates to be those which express intrinsic properties and relations, the answer is yes. Say that a formula containing free variables is “intrinsic” iff the result of turning it into a predicate, by lambda-abstracting each of its free variables, expresses an intrinsic property or relation. Conjunctions, disjunctions and negations of intrinsic formulae are themselves intrinsic. Moreover, when ψ is an intrinsic formula and v_1, v_2 are variables, $\exists_u v_1 (Iv_1 v_2 \wedge \psi)$ and $\forall_u v_1 (Iv_1 v_2 \rightarrow \psi)$ are intrinsic formulae: quantification over the parts of the given objects preserves intrinsicness. This is an extremely plausible principle about the notion of intrinsicness, and can easily be seen to be true on Lewis’s theory of intrinsic properties (as presented for example in Lewis 1983a). It follows that any formula in which x is the only free variable, in which all predicates express intrinsic properties and relations, and in which all quantifiers carry a restriction to the parts of x , is intrinsic. (If we understand ‘well-behaved’ more broadly, so as to apply got some non-intrinsic predicates provided that their application depends only on how an object is related to its Lewis-world-mates, we will very plausibly get the same result.)

The upshot of this discussion is that if we adopt *Restriction qua* analysis of modal claims involving some special kind of restricted quantification, there are no attractive alternatives to adopting *Generalized Redundancy* as our account of modal claims involving only unrestricted quantification.¹¹

¹¹Divers (1999) seems to agree that modal realists should accept *Generalized Redundancy*: ‘Whenever the possibility operator expresses a non-trivial semantic function on quantificational sentences it is, indeed, always that of altering the scope of formerly world-restricted quantifiers’ (229). The class of sentences on which Divers thinks the possibility operator operates only vacuously is, however, wider than the class of sentences built up using well-behaved predicates, unrestricted quantifiers and truth-functional operators. For Divers, the class also includes, for example, subject-predicate sentences where the subject is an object

3.3 Contingency-inducing quantification as *de re* restricted quantification

Suppose ϕ is a closed sentence built up from well-behaved predicates, truth-functional connectives, and quantifiers. According to *Generalized Redundancy*, $\Diamond\phi$ and $\Box\phi$ can differ in truth value from ϕ only when the quantifiers in ϕ are understood as tacitly restricted, somehow. But obviously not just any restriction on the quantifiers would suffice to introduce contingency. For example, if we interpret the quantifier in ‘Possibly, some swans are blue’ as tacitly restricted to *noisy* things, the sentence will be equivalent to ‘Possibly, some_u noisy swans are blue’, and hence to ‘some_u noisy swans are blue’, and hence to ‘some swans are blue’ on the same restricted interpretation. For a tacitly restricted quantifier to take us beyond the realm of the modally trivial, the restriction must not itself be expressible by some well-behaved predicate: otherwise, the claim made using the restricted quantifier could be made explicit in a language with only unrestricted quantifiers and well-behaved predicates. What, then, is the kind of restriction that makes for modal non-triviality?

One might suggest that the relevant restriction could be expressed by some predicate that is qualitative but ill-behaved. But it is hard to see how this could work. The usual examples of ill-behaved qualitative predicates are predicates for which we can provide, or at least gesture towards, analyses in terms of unrestricted quantifiers and well-behaved qualitative predicates—e.g. the analysis of ‘tallest person’ as ‘person taller than any other person’, or the analysis of ‘inventor of bifocals’ as ‘person who thought of bifocals before anyone else’. It was because we had such analyses vaguely in view that the restriction of Lewis’s schemas to sentences built up from well-behaved predicates didn’t seem like a terribly serious failure of generality. But if sentences built up from well-behaved predicates and unrestricted quantifiers are modally trivial, an ill-behaved predicate that took us beyond this realm into the realm of contingency could not be necessarily equivalent to any expression in a language with only well-behaved predicates.

What has just emerged is that, unless we posit some altogether new and mysterious category of qualitative but ill-behaved predicates, there is no way to accept *Generalized Redundancy* without accepting the following *prima facie* stronger principle, which we first encountered in section 3.1:

Modal Redundancy: When ϕ is a closed sentence built up entirely out of

that is not part of any Lewis-world. Section 5.2 will make it clear why I think this is a mistake.

qualitative vocabulary, $\Box\phi$ and $\Diamond\phi$ are equivalent to ϕ .

For unless the notion of a well-behaved predicate is restricted in some surprising way, it is extremely plausible that *every* purely qualitative proposition is necessarily equivalent to some proposition expressible in a language with only unrestricted quantifiers, truth-functional operators, and well-behaved predicates, or at least supervenient on such propositions.

I have been assuming, and will continue for now to assume, the standard view that ‘non-qualitative’ properties, relations and propositions are those that are not *de re* (‘singular’, ‘referential’, ‘haecceitistic’). Thus a non-qualitative restricted quantifier is a *de re* restricted quantifier—a quantifier restricted to items that are related in the right way to some particular thing or things. Such restrictions are common. For example, I might utter the sentence ‘there are no blue swans’ and thereby say, of a certain zoo, that there are no blue swans in it; or say, of the planet Earth, that there are no blue swans on it; or say that there are no blue swans on the same planet as *me*. These are all available readings for unembedded utterances of ‘there are no blue swans’, and surely they all remain available when this sentence is embedded under a modal operator. In saying ‘Possibly, there are no blue swans’, I might say that it is possible that there are no blue swans in a certain zoo; that it is possible there are no blue swans on Earth; or that it is possible that there are no blue swans on the same planet as *me*. A complete reductive account of modality should, in any case, tell us what it would be for each of those claims to be true. And there certainly are possible accounts of these claims which are consistent with *Generalized Redundancy*, but allow modal operators to be non-vacuous when combined with such *de re* sentences. BMCT, the simple version of counterpart theory with which we began, is one such analysis. It yields analyses such as the following:

It is possible that there are no blue swans in London Zoo := London Zoo has a counterpart in which there are no blue swans

It is possible that there are no blue swans on Earth := Earth has a counterpart on which there are no blue swans

It is possible that there are no blue swans on the same planet as *me* := one of my counterparts is such that there are no blue swans on the same planet as it

And as we saw in section 3.1, BMCT is not only consistent with *Generalized*

Redundancy: given some minimal logical assumptions, it *entails* the stronger principle of *Modal Redundancy*.

Lewis's

- (2) It is possible that no swans are blue := there is a Lewis world such that no swans in it are blue

does not look at all plausible as an account of what it is for it to be possible for there to be no blue swans *in London Zoo, or on Earth, or on the same planet as me*. Is there any way to interpret the quantifier on the left as carrying a *de re* restriction on which (2) and other instances of *Restriction* look more plausible? Yes: the obvious approach is to interpret the relevant quantifiers as carrying a *de re* restriction to things in a certain *Lewis-world*, the one of which we are parts. It will be convenient to have a name for this Lewis-world: let's call it 'Cosmo'. The suggestion, then, is that if one is determined to accept (2) on some interpretation or other, one should accept it on an interpretation that makes it equivalent to (5):

- (5) Possibly, no swans in Cosmo are blue := there is a Lewis-world such that no swans in it are blue.

More generally, the suggestion is that if we accept *Restriction* at all, we should accept it as a schema for analysing modal claims in a language each of whose quantifiers is equivalent to the result of adding the restriction 'in Cosmo' to an unrestricted quantifier.

How interested should we be in an analysis which only applies to modal claims whose quantifiers involve this particular kind of *de re* restricted quantifier, while saying nothing about any other kind of *de re* modal claim? It depends how widespread we take the practice of restricting quantifiers to things in Cosmo to be. If one thought that ordinary people believed in a plurality of Lewis-worlds, one might naturally be led to the hypothesis that our quantifiers are quite often restricted to things in Cosmo: this would perhaps help to explain why our belief in many Lewis-worlds doesn't show up more often in ordinary speech. But the idea that the folk believe in many Lewis-worlds strikes me as misguided, as it did Lewis. Honesty requires proponents of a Lewis-style ontology to recognise the radical character of their view, even if they want to go on to argue that it should be believed in spite of its radical character. Thus I see no reason to think that quantification in ordinary speech is often restricted just to the domain of things in

Cosmo. Typically, quantifiers are either unrestricted, or restricted to a much narrower domain than this one.¹²

Interesting or not, analyses like (5) are tenable (*modulo* the incredulous stare), at least on some resolution or other of the context-sensitivity of 'possible'. However, what we are ultimately looking for is a schema for analysing modal claims of all sorts. A theory which tells us what to do with sentences in which quantifiers are restricted to things in Cosmo, but says nothing about sentences in which quantifiers are restricted to things in London Zoo, on Earth, etc., is not very satisfactory. Moreover, it would be very desirable for this analysis to work in a *uniform* way, yielding an analysis of the same form for 'It is possible that no swans in *a* are blue' without having to check whether *a* names a zoo or a planet or a Lewis-world. We have already encountered an analysis that is uniform in this way, namely BMCT. BMCT can be provided to any modal claim containing exactly one singular term (and in which all other vocabulary is qualitative), no matter what sort of entity that term might designate. For example, it tells us that

- (6) Possibly, no swans in Cosmo are blue := there is a counterpart of Cosmo such that no swans in it are blue.

Prima facie, (5) and (6) look like competitors. But in fact it is not compulsory to think of them as such. We could make them consistent by giving an account of 'counterpart' according to which *the counterparts of any Lewis-world are all and only the Lewis-worlds*.¹³ In a context where 'counterpart' satisfies this constraint, BMCT entails the biconditional corresponding to (5). Indeed, it entails that this biconditional is *necessarily* true, thanks to the following instance of BMCT:

- (7) Necessarily, the counterparts of Cosmo are all and only the Lewis-worlds := every counterpart *x* of Cosmo is such that the counterparts of *x* are all and only the Lewis-worlds.

¹²Physics is the area of discourse in which it is most plausible to think that some quantifiers might be tacitly restricted to things in Cosmo. However, even here I think it is more plausible to think that absolutely unrestricted quantification is going on. Perhaps it is true that many physicists would claim to be talking only about Cosmo if we drew their attention to the fact that some deny that Cosmo is the whole of concrete reality. But in view of physicists' notorious tendency to start spouting positivistic slogans whenever a philosopher comes into the room, it would be dangerous to read much into this hedging behaviour.

¹³At at least one point Lewis suggests precisely this. At *OPW* 124, speaking of arbitrary worlds *W* and *V*, he says that 'the counterpart of *W* at *V* presumably is *V* itself'.

It is thus open to us to interpret *Restriction* not as part of a system of analyses which stands in competition with BMCT, but as nothing more than a special case of BMCT applying to sentences in which the only cases of *de re* reference occur in the form of a restrictor to things in a particular Lewis-world which attaches to every quantifier in the sentence.

Of course, this reconciliation is only successful to the extent that it is plausible to think that Lewis-worlds are one another's counterparts. Section 5.2 will consider whether this claim is well-motivated.

BMCT is not merely a useful concrete example of a theory that allows for non-trivial *de re* modal claims while being consistent with *Modal Redundancy*. Bearing in mind that 'counterpart' is a black box which we can fill in as we see fit, once we accept *Modal Redundancy* there is no real alternative to accepting BMCT for singly *de re* modal claims. Provided that we allow ourselves to engage in quantification over qualitative properties in the obvious way, if we accept *Modal Redundancy*, there is bound to be an interpretation of 'counterpart' on which BMCT can be endorsed, namely one on which ' x is a counterpart of y ' is taken to mean 'the complete qualitative profile of x is such that possibly, y instantiates it'. Under this interpretation, BMCT's analysis for ' $\Diamond\phi(\alpha)$ ' is equivalent to the following claim:

- (8) $\Diamond\phi(\alpha) :=$ there is some object x such that $\phi(x)$ and x 's complete qualitative profile is such that possibly, α instantiates it

The right-to-left direction of (8) is straightforward. Since we are requiring that all vocabulary in $\phi(\alpha)$ except for the term α is purely qualitative, the question whether an object satisfies $\phi(x)$ is determined by that object's complete qualitative profile. If some object x such that $\phi(x)$ instantiates a certain qualitative profile, then instantiation of that profile must *entail* satisfaction of ' $\phi(x)$ '; hence if it is possible that α instantiate that profile, it must be possible that α itself should satisfy ' $\phi(x)$ '. For the left-to-right direction of (8), suppose that it is possible that $\phi(\alpha)$. Then since it is necessary that α instantiate some qualitative profile or other, and qualitative properties are necessary existents, there must be some complete qualitative profile Q such that possibly ($\phi(\alpha)$ and α instantiates Q). But since the question whether $\phi(x)$ for some given x depends only on x 's qualitative character, any such Q must be such that necessarily, whenever some object x instantiates Q , $\phi(x)$. Now, if it is possible that α instantiates Q , it is of course possible that *something* instantiates Q .¹⁴ By *Modal Redundancy*, this entails that something *does* instantiate Q :

¹⁴One might want to resist the argument at this point for some special cases, where

being purely qualitative, the claim that something instantiates Q is necessary if true, and impossible if false. Thus there is some object x such that Q is x 's complete qualitative profile, such that $\phi(x)$. Since Q is a qualitative profile that α could instantiate, this establishes the right-hand side of (8).

Let me recapitulate the dialectic of the last two sections. Suppose that you think that analyses like Lewis's

- (2) It is possible that no swans are blue := there is a Lewis world such that no swans in it are blue

are true on some interpretation or other. If so, you should take the relevant interpretation to be one involving some kind of restricted quantification on the left hand side, otherwise you will run afoul of Puzzles 1–3. You will need some other account of the meaning of modal operators applied to sentences involving unrestricted quantifiers. There are essentially only two options available: the S5-rejecting, 'appropriate part of reality' analysis, and *Generalized Redundancy*, according to which sentences built out of unrestricted quantifiers and well-behaved predicates are modally trivial. The latter is more attractive. But if the standard examples of ill-behaved predicates—'tallest person', 'inventor of bifocals', etc.—are representative of the class, acceptance of *Generalized Redundancy* commits us to acceptance of *Modal Redundancy*, according to which *all* purely qualitative closed sentences are modally trivial—all contingency is *de re* contingency. But if you think this, your general account of *singly de re* modal claims must be BMCT: to be possibly a certain qualitative way is to be appropriately related to something which is that way (i.e. to have a "counterpart" which is that way). If you still really want to hold on to claims like (2), you can do so by interpreting the quantifier on the left as restricted to things in Cosmo, and accepting that the

the truth of $\phi(\alpha)$ is compatible with α 's failure to exist. If we are prepared to speak of complete qualitative profiles which entail *non-existence*, so that for each thing, necessarily, if it instantiates that qualitative profile it fails to exist, we will think that when Q is one of these special qualitative profiles, it is possible that α instantiate Q but not possible that something instantiate Q . Alternatively, we could claim that things cannot instantiate any qualitative profiles without existing: in that case, when $\phi(\alpha)$ is compatible with α 's failure to exist, we will resist the argument from 'possibly $\phi(\alpha)$ ' to 'there is some Q such that possibly ($\phi(\alpha)$ and α instantiates Q)'. However, these ways of resisting the argument are of limited interest, insofar as we still get BMCT for the wide variety of choices of $\phi(\alpha)$ which are not by anyone's lights consistent with ' α does not exist'. I will return to the question what counterpart theorists should think about possible non-existence in section 3.6.

counterparts of each Lewis-world are all and only the Lewis-worlds. But it is unclear what would motivate this claim.

People often think of counterpart theory as a view which isn't really *serious* about *de re* modality. They think of the view as driven by the idea that the *interesting* modal facts are *de dicto* modal facts—natural languages allow for *de re* modal claims, and so we need to give some theory of their truth conditions, but from the point of view of the metaphysician, this is a bit of a distraction—it would be more perspicuous if we all spoke a language in which *de re* modal claims couldn't be formulated at all. But if we develop our counterpart theory in a way that commits us to *Modal Redundancy*, this is all backwards. The theory of *de re* modality is all there is to the theory of modality; until we start to deal with *de re* claims, the modal facts are completely trivial and uninteresting. Lewis's way of developing counterpart theory initially seemed to give us a way of avoiding *Modal Redundancy*, and preserving an interesting role for non-*de re* modality. But, as we discovered, this was a mirage; we run into horrible trouble unless we interpret Lewis's theory as a theory of restricted quantification, and it is hard to see what the relevant restriction could be if isn't a *de re* restriction; but if it is a *de re* restriction, Lewis's theory does not provide us with the non-trivialising account of non-*de re* modality we thought we wanted.

There is a parallel dialectic in the temporal case. Suppose that you believe that spacetime can be partitioned into special three-dimensional regions—call them “Slices”—in such a way that each ordinary object is, in some way we need not specify at this stage, intimately associated with, or ‘in’, exactly one Slice. You seek a reductive theory of temporal operators. You might be tempted by the thought that whatever you end up saying about *de re* temporal claims, *de dicto* temporal claims can be handled by the temporal analogues of *Restriction*:

$$\begin{aligned} (Restriction_T) \quad F\phi &:= \exists s(Fs \wedge [\phi]^s) \\ P\phi &:= \exists s(Ps \wedge [\phi]^s) \end{aligned}$$

where ‘*Fs*’ and ‘*Ps*’ might be pronounced as something like ‘*s* is a futuristically-accessible Slice’ and ‘*s* is a historically-accessible Slice’. For example,

- (9) It has been the case that some swans are hungry := there is a pastly-accessible Slice some swans in which are hungry.
- (10) It will be the case that no swans are hungry := there is a futurely-

accessible Slice no swans in which are hungry.

But you should not accept such analyses on an unrestricted reading of the quantifiers on the left, since doing so would have implausible results every bit as bad as those that emerged as Puzzles 1–3. Since ‘there once were_u hungry swans’ entails ‘there are_u hungry swans’, we cannot endorse the claim that it will be the case that (there are no_u hungry swans, but there once were_u hungry swans), despite the fact that this follows (intuitively, and in standard tense logic) from the premise that there are_u hungry swans although there will not always be_u hungry swans. In fact, if we think that there will at some point be no hungry swans, we will be forced to say that it will also someday be the case that *there have never been*_u hungry swans. Moreover, even though we think that there are two things which belong to different Slices, we will have to claim that it never will be the case, and never has been the case, that there are two such things.¹⁵

If you want to accept instances of (*Restriction_T*) like (9) and (10), you had better interpret the quantifiers on the left as somehow restricted. But then you will need to give some other theory about what temporal operators mean when combined with unrestricted quantifiers. There is no tenable alternative to accepting the temporal analogue of *Generalized Redundancy*. (The temporal analogues of the ‘appropriate part of reality’ analysis are completely untenable, since the temporal analogue of the conflict with S5 which emerged in the modal case is a conflict with the mixing principles—what is the case always will have been the case; what is the case has always been going to be the case—which are absolutely basic to the logic of the temporal operators.) And just as in the modal case, unless you are willing to posit some mysterious ill-behaved yet qualitative predicate that works very differently from any ordinary examples, this means you will have to accept *Temporal Redundancy*: temporal operators are redundant when applied to purely qualitative sentences; all change is *de re* change. But if you embrace this claim, your account of *singly de re* temporal claims must be BTCT: to be going to be [to have been] a certain qualitative way is to have a future-counterpart [past-counterpart] which is that way. If you still really want to

¹⁵The badness of this result is not *quite* so obvious as the badness of violating T in the modal case. Some conceive of the notion of a ‘timeless truth’ in such a way that when it is a timeless truth that ϕ , ‘it is sometimes the case that ϕ ’ is false. I think this is a mistake, born of a confusion between infelicity due to the presence of a vacuous element and falsehood. No matter what the subject matter, it is intolerable to assert ϕ while also asserting ‘People who assert that ϕ never speak truly’.

hold on to (*Restriction_T*), you can do so by interpreting the quantifiers on the left as restricted to things in the Slice to which *we* belong, and accepting that Slices only have other Slices as future- and past-counterparts. But it remains to be seen whether this is something you should want to say.

The most interesting way to resist these arguments is to reject the step from *Generalized Redundancy* (and its temporal analogue) to *Modal Redundancy* (and *Temporal Redundancy*). We could posit some special predicate-meaning, which is not *de re*, but also not equivalent to anything expressible in terms of well-behaved predicates, and take the presence of this meaning, either as the meaning of an explicit predicate or as a tacit quantifier restriction, to be a source of contingency quite different from *de re* contingency. If we wanted, we could then endorse *Restriction* and (*Restriction_T*) on a reading where the quantifiers on the left hand side are taken to be restricted in the special way, while those on the right are unrestricted. Assuming that we want to allow for *de re* contingency [change] in any case, it is not clear what the motivation for such a posit would be; nevertheless, some philosophers seem to take it seriously. So, section 3.5 will be devoted to exploring some forms such an account might take, and showing they are deeply problematic. But first. . .

3.4 An exegetical interlude

To what extent did Lewis actually accept the views I have been offering him? Did he, for example, accept that modal operators are redundant when applied to purely qualitative sentences?

There is considerable evidence that suggests that he did. For one thing, Lewis does not seem to embrace the puzzling claims that—as we saw in section 3.2—follow from *Restriction* when all the quantifiers are taken as unrestricted. There are, moreover, several passages in which Lewis directly considers the modal status of claims involving unrestricted quantification, and seems to suggest that they are necessary when true, i.e. that *Generalized Redundancy* is true. Here is one example:

Our *contingent* knowledge that there are donkeys at *our* world requires causal acquaintance with the donkeys, or at least with what produces them. Our *necessary* knowledge that there are donkeys at *some* worlds—even talking donkeys, donkeys with dragons as worldmates, and what have you—does not require causal acquaintance either with the donkeys or with what produces them. (OPW 112)

This presupposes that it is necessary that there are donkeys at some worlds, and thus that it is necessary that there are donkeys, unrestrictedly speaking. (It also suggests, interestingly, that the true reading of 'it is contingent that there are donkeys' is equivalent to 'it is contingent that there are donkeys at our world', which seems to be a *de re* modal claim about a particular Lewis-world, or perhaps about *us*: this does something to support the attribution to Lewis of *Modal Redundancy*.) In a similar vein, Lewis says that 'reality might have been different' is not true, if we understand 'reality' in such a way as to make 'reality is the totality of everything' true (OPW 101, note 1). And in Lewis 1996, he accepts without demur the attribution to him of a theory that 'treat[s] all statements about modal reality as non-contingent: if any such statement is possibly true, then it is true *simpliciter*'.¹⁶

Also supporting the attribution of *Modal Redundancy* to Lewis are several passages in which he suggests that the accessibility relation on worlds and the counterpart relation on proper parts of worlds are just two subcases of a more general relation of one thing being a 'possibility for' another:

As other worlds are alternative possibilities for an entire world, so the parts of other worlds are alternative possibilities for lesser individuals. . . . As quantification over possible worlds is commonly restricted by accessibility relations, so quantification over possible individuals is commonly restricted by counterpart relations. (OPW 8)

But I say that *any* possible individual is a possibility, and not all possible individuals are possible worlds. Only the biggest ones are. . . . [A possible world] is a way that an entire world might possibly be. But lesser possible individuals, inhabitants of worlds, proper parts of worlds, are possibilities too. They are ways that something less than an entire world might possibly be. (OPW 230)

These claims tend to suggest that the most perspicuous and general approach to the analysis of modal claims would be expressed in terms of the generic

¹⁶Another relevant passage: 'If I had acted otherwise, for instance by taking to a life of crime, each and every good or evil that is present somewhere in the totality of worlds would still have been present, and none would have been added. It is wrong to think: then this world would have been a little worse, and the rest would have been no different, so the totality of worlds would have been a little worse.' (OPW 123) Given that Lewis endorsed the inference from 'necessarily ϕ ' to 'if it were that ψ , then it would be that ϕ ', if he had endorsed *Restriction* as an analysis of possibility claims embedding unrestricted quantifiers, he should instead have said 'if I had acted otherwise there would have been only one Lewis-world'. (See chapter 5 for more discussion of this passage.)

'possibility for' relation, and would not require singling out Lewis-worlds for any special treatment.

On the other hand, the first few pages of *OPW*, in which the role of Lewis-worlds in the analysis of modal claims is introduced, strongly suggest a view on which *Generalized Redundancy* fails. We are told that 'possibly there are blue swans iff, for some world *W*, at *W* there are blue swans', where 'at *W*' 'works mainly by restricting the domains of quantifiers in its scope, in much the same way that the restricting modifier "in Australia" does' (*OPW* 5). It would be odd to think that 'In Australia, all swans are black' expresses a truth only when we interpret the quantifier as carrying some tacit restriction: since we have gone to the trouble of making 'in Australia' explicit, there is no obvious need to posit any further implicit restriction. On Lewis's account of 'at *W*', then, it looks like it can apply restrictions *even when it is applied to a sentence that doesn't come with its own tacit restriction*, so that 'at *W*, Q_u *Fs* are *Gs*' means the same as ' Q_u *Fs* which are in *W* are *Gs*'. If this is right, 'At some Lewis-world *W*, no swans are blue', and hence also 'Possibly no swans are blue', will be true even when the quantifier is just the familiar unrestricted quantifier.

There is, moreover, some textual evidence that suggests that the restrictions apply even to unrestricted quantifiers, for example the following passage:

We wanted to ask whether two worlds could differ in their laws without differing in their distribution of local qualitative character. But if we read the 'could' as a diamond, the thesis in question turns into this: it is not the case that, possibly, two worlds differ in their laws without differing in their distribution of local qualitative character. In other words: there is no world wherein two worlds differ in their laws without differing in their distribution of local qualitative character. That's trivial—there is no world wherein two worlds do anything. At any one world *W*, there is only the one single world *W*. The sentential modal operator disastrously restricts the quantification over worlds that lies within its scope. (*OPW* 16)

Admittedly, it is not clear that this passage commits Lewis to the claim that the restrictor analysis works for modal claims expressed using the English word 'possibly': the point of the passage is to argue that certain claims in English are not well expressed in a certain *formal* language. But Lewis certainly doesn't suggest that he has up his sleeve a *better* analysis of 'pos-

sibly' in English which allows it to behave differently from 'diamond' in the relevant respect.¹⁷

Some remarks of Lewis's suggest an alternative strategy for blocking the reasoning behind the puzzles without embracing *Generalized Redundancy*. For his treatment of 'at W' includes an important qualification, which I left out in my earlier presentation of the view:

I do not suppose that [restrictive modifiers] must restrict all quantifiers in their scope, without exception. 'In Australia, there is a yacht faster than any other' would mean less than it does if the modifier restricted both quantifiers rather than just the first. . . . [A] lot is left up to the pragmatic rule that what is said should be interpreted so as to be sensible. (OPW 6).

Thus according to Lewis, sentences involving restricting modifiers, including modal operators, admit a range of readings, which differ as regards which quantifiers end up having restrictions attached to them. So, for example, 'In W there are many Lewis-worlds' and 'Possibly there are many Lewis-worlds' have readings on which they are both equivalent to 'There are many Lewis-worlds', as well as readings on which they are trivially false. And it is easy to imagine a pragmatic story about why we would tend to prefer the former readings.

The profusion of readings generated by making the quantifier-restricting effects of modal operators optional makes the puzzles from section 3.2 harder to raise, by making the application of logic within the scope of modal operators a complex affair. For example, a sentence like

- (11) Possibly (there are no blue swans and there is a Lewis world that contains at least one blue swan)

¹⁷At OPW 211, Lewis sounds like he is rejecting T, at least for some notion of possibility: 'I do not deny the existence of trans-world individuals, and yet there is a sense in which I say that they cannot possibly exist.' But this is complicated by the immediately following explanation: 'As should be expected, the sense in question involves restricted quantification.' This remark implicates that there is another sense in which 'trans-world individuals cannot possibly exist' is *false*, and that this sense involves *unrestricted* quantification. This looks like good news for the attribution of the redundancy thesis to Lewis. However, it is not entirely clear that Lewis is attributing the difference between the two readings of 'trans-world individuals cannot possibly exist' to a variation in the interpretation of the quantification introduced by 'exist'—it might be suggested that the sense in which the true reading of 'transworld individuals cannot possibly exist' 'involves restricted quantification' is just that restricted quantifiers will crop up in the *analysis* of this claim.

can be read as expressing a truth (by restricting only the first quantifier), despite the logical inconsistency of the sentence to which the operator is applied. Of course, those who accept *Generalized Redundancy* can also allow for true readings of (11), in which the first quantifier is tacitly restricted and the others are not; but on this view, the true readings of (11) correspond to readings of the embedded sentence on which it is not *really* logically inconsistent. If we want to interpret Lewis's idea of optional restriction as a genuine alternative to *Generalized Redundancy*, we cannot say this. This is a heavy cost: many uses of modal operators would be seriously compromised if we cannot reason from the logical inconsistency of a sentence on a certain resolution of its context-sensitivity to the falsehood of the result of prefixing that sentence with 'Possibly' while holding fixed the interpretation of the context-sensitive elements in the original sentence. Moreover, while the approach allows us to avoid having to make certain bad-sounding speeches, it does not straightforwardly allow us to claim validity for modal principles like S5 and T. As I understand it, the claim that T is *valid* (for metaphysical necessity, say) should commit us to the claim that for every sentence ϕ , $\ulcorner \phi \rightarrow (\text{it is metaphysically possible that } \phi) \urcorner$ is true on *all* admissible readings, provided that any ambiguity or context-sensitivity in ϕ is resolved in the same way in both occurrences. If we think that T is valid, then, there is no way to avoid claiming that the true readings of 'There are at least two Lewis-worlds, and it is not metaphysically possible that there are at least two Lewis-worlds' involve differential resolution of some context-sensitive constituent within the sentence 'there are at least two Lewis-worlds' itself. And the only plausible source of relevant context-sensitivity in *this* sentence is the one associated with quantifier domain restriction.

3.5 Alternative accounts of contingency-inducing quantification

To accept *Generalized Redundancy* without accepting *Modal Redundancy*, one must posit a way of interpreting quantifiers as carrying some restriction which is neither not *de re*, but also not equivalent to anything expressible in terms of well-behaved qualitative predicates. How could this work?

On one version of the view, the special kind of interpretation is available *only* when quantifiers are embedded under modal [temporal] operators. In this respect, it is like the bound interpretation of pronouns, which is available for pronouns in the scope of appropriate quantifiers, but corresponds to no reading that unembedded pronouns can bear. Semantically, it would be

natural to implement this view by thinking of the restrictor as involving a hidden variable, which is bound by the relevant modal [temporal] operator. This is reminiscent of the kind of story told by B-theorists. But the account of modal [temporal] operators required by this view is much more complicated than the standard B-theoretic conception of them as quantifiers over worlds [times]. For given counterpart theory, these operators are not *just* quantifiers, since they can have a non-redundant effect even when they are attached to a sentence which, like 'Obama is tall', contains no variables for them to bind.

On this view, there are two structurally different ways to get a true reading for the sentence 'It could have been the case that no swans were white'. One possibility is to posit a *de re* restriction in 'no swans'. Syntactically, this could take the form of an unpronounced constituent 'in x ', where the variable x is free and interpreted deictically. On this reading, our metaphysical account of what it is for the sentence to be true will involve quantification over counterparts of x . A quite different possibility is to posit an unpronounced constituent 'in x ', where x is *bound* by the operator 'it could have been the case that'. On this reading, the counterpart relation is irrelevant.

The main objection to this view is its needless, and arbitrary, complexity. It is one thing to suggest, as modal B-theorists do, that modal operators are *always* variable-binders. It is another thing to suggest that they *sometimes* bind hidden variables, and sometimes apply non-vacuously to closed sentences (like 'Obama is tall') by introducing quantifiers over counterparts. There is no linguistic motivation for drawing a distinction in syntactic structure between 'Possibly, Obama is not tall' and some uses of 'Possibly, no swans are white', according to which variable-binding takes place only in the latter case. Even more obviously, there is no linguistic motivation for positing two different syntactic structures for 'Possibly, no swans are white'. (How are we supposed to tell which of the two mechanisms is at work in any given case?) I am not saying, here, that our ultimate *metaphysical* account of what it is for it to be possible or necessary that P has to work in any particularly uniform way; only that metaphysical considerations should not drive us to posit differences of the sort *syntax* cares about, such as differences in variable-binding structure, unless we can motivate those differences using linguistic as opposed to metaphysical considerations.

Let us assume, then, that the distinctive kind of restricted interpretation—contingency-inducing, yet not *de re*—is available even for unembedded uses of quantifiers. I will continue to use the subscript r to indicate the relevant form of restriction. For the sake of having a concrete example, let us suppose

that no_r swans are blue, even though some_u swans are blue. Here is the question I want to press. Consider a person, Twin, who is generally similar to me as regards his intrinsic and environmental properties, but who happens to be part of a Lewis-world—Cosmo-Prime—which also contains blue swans. Suppose I say ‘some swans are blue’, and thereby say (falsely) that some_r swans are blue. When Twin says ‘some swans are blue’ under analogous circumstances, does he also say that some_r swans are blue?

There is a compelling argument that he does not. We have assumed that

(P1) No_r swans are blue.

From the truth schema (see section 1.2), we know that

(P2) The proposition that some_r swans are blue is true iff some_r swans are blue.

But it seems obvious that

(P3) When Twin says ‘some swans are blue’, everything he says is true.

When people in Cosmo-Prime see blue swans swimming around and say ‘some swans are blue!’, the belief they thereby express is (usually) paradigmatically justified by ordinary perceptual methods. Are we supposed to think of Cosmo-Prime as a place of epistemic tragedy, where following these excellent methods leads people to form systematically untrue beliefs? Surely not; hence (P3). But from (P1), (P2), and (P3), elementary logic yields

(C) When Twin says ‘some swans are blue’, he doesn’t say that some_r swans are blue.

If one follows this kind of reasoning where it leads, one will end up thinking that the practice of using ‘some swans are blue’ to say that some_r swans are blue is more or less confined to Cosmo. People in other Lewis-worlds, even ones in most respects similar to Cosmo, do not express the very same propositions as us when they say the words that in our mouths express claims about what there is_r.

(It would be a bad mistake to confuse (C) with the counterfactual claim that if I had been spatiotemporally related to blue swans but otherwise qualitatively similar to the way I actually am, *I* would not have used ‘some swans are blue’ to say that some_r swans are blue. There are lots of things that people in other Lewis-worlds never do that I would have done even if I had

been qualitatively different in all kinds of ways. For example, no-one in any other Lewis-world lives on Earth. But my living on Earth is counterfactually robust: there is a wide range of qualitative properties such that I would still live on Earth even if I had them. For example, I would still live on Earth if I were a philosopher of mind. Chapter 4 below will discuss how counterpart theorists should analyse “multiply *de re*” modal claims like ‘I could live on Earth while being a philosopher of herpetology’; but it is a constraint on the adequacy of such analyses that they be compatible with the fact that no philosophers of herpetology live on *Earth*.)

(C) would not be surprising iff we were taking the quantifiers subscripted with ‘r’ to be some kind of *de re* restricted quantifiers. On this view, to say that some_r swans are blue is to say that some swans on Earth are blue, or that some swans in Cosmo are blue, or some such thing. And of course, this isn’t the sort of thing we should expect to find people at other Lewis-worlds doing very often. For one thing, saying that some swans on Earth are blue seems to involve *referring* to Earth; and it is not usually thought to be easy to refer to objects towards which one bears no spatiotemporal or causal relations.¹⁸ When I say that some swans on Earth are blue, Twin surely does not: rather, he says that some swans on Earth-Prime are blue, where Earth-Prime is the planet Twin lives on. Similarly, when I say that some swans in Cosmo are blue, Twin under analogous circumstances says, instead, that some swans in Cosmo-Prime are blue. But if, as we are assuming, the claim that some_r swans are blue does *not* involve *de re* reference to Cosmo or any of its parts, it becomes hard to see how (C) could be true. How could the mere fact that Twin and I are parts of different Lewis-worlds be relevant to the facts about which propositions we express, other than *de re* propositions about ourselves or our surroundings? It is unmysterious how a *de re* proposition could be much easier to express for people in one Lewis-world than for people in another, very similar Lewis-world; it is very mysterious what could make for such differences in the case of a non-*de re* proposition.

The “pictorial ersatzism” Lewis argues against in *OPW* §3.3 is a view

¹⁸Perhaps there are some contexts (see Hawthorne and Manley MS, ***) in which reference is very easily achieved—maybe it is even true in some of these contexts that whenever one says that all *F*s are *G*s, and *x* is an *F*, one says that *x* is *G*. By these lax standards, Twin can easily say that some swans on Earth are blue, just by saying that some swans on every planet (unrestrictedly speaking) are blue. But it is implausible that all contexts work like this; we can run our argument within a less permissive context. And even by very permissive standards, it is still not plausible that Twin says that some swans on Earth are blue when he utters the sentence ‘some swans on Earth are blue’.

on which (P3) and (C) both fail. There is a special property “vim”, which we often use in restricting our quantifiers. Cosmo and its parts have vim; other Lewis-worlds don’t. Vim could have been distributed differently with respect to ordinary qualitative properties—for example, there could have been a Lewis-world with vim and containing blue swans. ‘Vim’ is not a well-behaved predicate: *Restriction* and RCT cannot be applied when we are dealing with sentences in which ‘vim’ occurs in the scope of a modal operator. For example, there could have been blue swans with vim even though no Lewis-worlds contain blue swans with vim. One doesn’t need to have vim to be a swan, to be a person, to be conscious, etc: thus predicates like ‘swan’, ‘person’, and ‘conscious’ can still be treated as well-behaved for the purposes of applying *Restriction*. It is reasonable for us to be highly confident that we have vim. It is also reasonable for the many, many people who lack vim to be highly confident that they have vim. But they are just wrong. The vimless part of reality is a place of pervasive epistemic tragedy, where good reasoning leads people to form systematically false beliefs about the distribution of vim.¹⁹

This is not an attractive view. Applying the epistemological standards that normally apply to the postulation of new primitive entities or properties, there seems to be no justification for postulating vim at all, or—assuming that we do postulate it—for believing anything very specific about its distribution. Of course, it is part of the view that these are the wrong standards to

¹⁹Phillip Bricker’s view (2001, 2006) is similar in many ways to the one just described. There are two unimportant differences: he calls vim ‘actuality’, and instead of saying that it is a qualitative but ill-behaved property, he says that it is non-qualitative, thereby departing from the standard usage in which ‘non-qualitative’ is equivalent to ‘*de re*’ or ‘*haecceitistic*’. There is one moderately important difference: unlike Lewis’s pictorial ersatzist, Bricker thinks it is possible that more than one Lewis-world has vim—indeed, he thinks that any fusion of Lewis-worlds could have vim. And there is one really important difference: according to Bricker, people who lack vim do not refer to vim, or at least do not do so when their ordinary qualitative properties are of such a nature that a vim-endowed person with those properties would be referring to vim. On his preferred version of the view, when we express propositions about the distribution of vim, qualitatively similar but vimless people fail to express anything at all. The vimless part of reality is, in a sense, a place of epistemic tragedy; but the tragedy is not that good reasoning leads to false beliefs, but that it leads to contentless attempts at thought. (C) is thus *true*, assuming that Twin lacks vim.

Bricker’s central argument for going down this route is his rejection of Lewis’s doctrine of the indexicality of actuality. In section 4.6, I will argue that my favourite form of modal realism—one that accepts BMCT and *Modal Redundancy*—should also jettison this part of Lewis’s view. Thus Bricker’s arguments do not provide reason for preferring his view over my favourite view.

apply, since the epistemological standards governing beliefs about vim are quite different from the standards governing beliefs about any other property. But an implausible metaphysics is not generally made better by being combined with an implausible epistemology, one tenet of which is that the metaphysics is something it is reasonable to believe. Things might be different if striking theoretical simplifications could be achieved by taking vim to be involved in the analysis of our ordinary beliefs; but it is obscure what the theoretical benefits are supposed to be.

One important motivation for wanting there to be a reading of 'no_r swans are blue' on which (C) fails is the doctrine, defended by Lewis (*OPW* 1.4), that propositions—at least in one central sense—are sets of Lewis-worlds. Anyone who accepts this will presumably think that there is a proposition, call it *p*, which is identical to the set of Lewis-worlds that contain blue swans. *p* looks like a natural candidate to be the proposition that some_r swans are blue. But if we sometimes assert *p* by saying 'some swans are blue', surely people in other Lewis-worlds can do so with equal ease. There is nothing in the nature of *p* that could make it easier to express here in Cosmo than anywhere else, given any sensible theory of what it takes to assert a set of worlds.

According to Lewis, not only some but *all* propositions can be identified with sets of Lewis-worlds. I think this is a clear mistake. Given that Earth is part of just one Lewis-world, there is just no plausible way to identify, say, the proposition *that no swans on Earth are blue* with any set of Lewis-worlds. For if this proposition is a set of worlds, which set of worlds could it be? If we help ourselves to a counterpart relation, there are a few natural candidates: the set of Lewis-worlds in which *some* counterpart of Earth lacks blue swans; the set of Lewis-worlds in which *every* counterpart of Earth lacks blue swans; the intersection of these two sets; the set of Lewis-worlds in which there is exactly one counterpart of Earth, which lacks blue swans. But any suggestion of this form is deeply problematic, for several different reasons.

- (i) 'Counterpart' is context-sensitive; we will get different sets of worlds depending on how we understand it. But while it is arguable that there is some context-sensitivity in the sentence 'No swans on Earth are blue' (e.g. because of shiftiness as regards what counts as 'blue'), it is not context-sensitive in the same way as 'counterpart'; and we could

have used a sentence ('Particle *a* is negatively charged') which is not plausibly context-sensitive at all. So the use of any particular counterpart relation as the basis for identifying a particular set of worlds as the proposition that there are no blue swans on Earth is objectionably arbitrary.

- (ii) Whichever candidate we choose, we will be committed to the claim that whenever the counterparts of *x* are exactly the counterparts of Earth, the proposition that there are no blue swans on *x* is identical to the proposition that there are no blue swans on Earth. So if Earth-Prime and Earth happen to be similar enough to share exactly the same counterparts, the proposition that no swans on Earth-Prime are blue is the proposition that no swans on Earth are blue.²⁰ But on any ordinary account of the counterpart relation, there is nothing to stop this from happening even when there *are* blue swans on Earth-Prime but not on Earth. So we get a violation of the truth schema: there are blue swans on Earth-Prime, and there are no blue swans on Earth, and yet the proposition that there are blue swans on Earth-Prime is the proposition that there are no blue swans on Earth.
- (iii) We also get trouble with propositional attitudes. If the proposition that there are blue swans on Earth-Prime is the proposition that there are blue swans on Earth, then anyone who believes or asserts that there are blue swans on Earth-Prime also believes or asserts that there are blue swans on Earth. And this is just incredible. People on Earth-Prime may very well believe, and assert, that there are blue swans on Earth-Prime; they obviously do *not* believe or assert that there are blue swans on *Earth*, a planet to which they don't bear any spatiotemporal relations, let alone the kinds of causal relations sometimes thought necessary for reference to occur. They may very well not believe or assert *anything* false.
- (iv) According to Lewis, some objects have multiple counterparts within a single Lewis-world. Suppose that Earth has two counterparts in some Lewis-world *w*, exactly one of which is inhabited by blue swans. If the

²⁰If we endorsed the claim (entertained in section 3.3) that each Lewis-world has all and only the Lewis-worlds as counterparts, then we will get an especially pernicious form of this problem: for any two Lewis-worlds *w* and *v*, the proposition that there are no blue swans in *w* will be identical to the proposition that there are no blue swans in *v*.

proposition that there are blue swans on Earth is a set of Lewis-worlds, it either does or doesn't contain w . In either case, we are in danger of having to accept something obviously false. If the proposition does contain w , it looks like 'The proposition that there are blue swans on Earth is such that possibly, it is true even though there are no blue swans on Earth' will come out true. If it doesn't, it looks like 'The proposition that there are blue swans on Earth is such that possibly, it isn't true even though there are blue swans on Earth' will come out true. At least, this is how things go if we adopt a Lewis-style counterpart theory in which modal operators introduce quantification over Lewis-worlds, and analyse 'true' in a modal context as 'contains w ', where w is a variable bound by the relevant quantifier. And if we are not planning to understand ascriptions of truth in terms of membership in this way, the identification of propositions with sets of worlds loses most of its point.²¹

Given counterpart theory, sets of Lewis-worlds are just terrible as candidates to be identified with *de re* propositions, such as the proposition that there are blue swans on Earth. If we want to identify these propositions with set-theoretic constructions of some sort, we will do much better to employ the kind of construction often used in the literature on 'structured propositions'. For example, we might take the proposition that there are blue swans on Earth to be the ordered pair ⟨Earth, the set of things on which there are blue swans⟩, or perhaps some more complicated construct that more closely mirrors the syntactic structure of the sentence 'there are blue swans on Earth'. (Even Lewis was willing to grant that such n -tuples could reasonably be identified with propositions, on one legitimate conception of propositions: see *OPW* 57–9.) So long as we have some counterpart-theoretic account of the multiply *de re* which lets us endorse claims like 'Earth's singleton set is such that necessarily, Earth is a member of it', we will then be able to endorse consequences of the truth schema like 'The proposition that there are blue swans on Earth is such that necessarily, it is true iff there are blue swans on Earth'. We can do by saying that a proposition that is an ordered pair of an object and a set is "true" iff the object belongs to the set in question, and extending this appropriately to propositions with more complicated kinds of structure.

* * *

²¹Dorr 2005 develops this point in much more detail.

So much for the claim that *all* propositions are sets of Lewis-worlds. Someone might, however, accept my arguments against this claim while insisting that it is still true that *some* propositions are sets of Lewis-worlds. It is not clear what could motivate such a view: Lewis's own argument for taking propositions to be sets of Lewis-worlds turns on systematic considerations which would have to apply across the board if they worked at all. Still, it is worth thinking through the consequences of a view which says that there are *some* propositions which can be identified with sets of Lewis-worlds, and in particular takes the set of blue-swan-containing Lewis-worlds to be the proposition expressed by 'some swans are blue' on the reading where it is contingent but not *de re*.

Proponents of such a view must find a way to resist the argument for (C) presented in section 3.5. Their response will surely turn, somehow, on a piece of ideology that the identification of propositions with sets of Lewis-worlds makes available, namely the notion of a proposition's being "true at" a Lewis-world: a proposition that is a set of Lewis-worlds is true at *w* iff *w* is one of its members. But how is the availability of this relativised truth predicate supposed to help us to resist the argument for (C), whose premises used monadic 'true', not 'true at'?

One thought is that we should be *contextualists* about monadic 'true' (at least when applied to propositions that are sets of Lewis-worlds). According to this thought, 'true' is context-sensitive: a given occasion of utterance, it is equivalent to 'true at *w*', where *w* is some conversationally relevant Lewis-world. By default, the Lewis-world relevant to an utterance is the one the speaker is part of. This default plausibly applies to a sentence like (P2), which as uttered by us will normally express the same thing as (P2'):

(P2) The proposition that some_r swans are blue is true iff some_r swans are blue.

(P2') The proposition that some_r swans are blue is true *at Cosmo* iff some_r swans are blue.

However, it is open to the contextualist to allow exceptions to the rule that the speaker's Lewis-world is always the relevant one, in cases where other Lewis-worlds have been made salient. By allowing such exceptions, the contextualist can account for the apparent truth of (P3), by claiming that in context, it is naturally interpreted as equivalent to (P3'):

(P3) When Twin says 'some swans are blue', everything he says is true.

(P3') When Twin says 'some swans are blue', everything he says is true *at Cosmo-Prime*.

Thus (P2) and (P3) are both true if we resolve their context sensitivity in the most natural way; but in that case the argument to (C) is invalid. If we resolve the context-sensitivity of 'true' uniformly, the argument is valid but unsound: if we understand 'true' as 'true at Cosmo', (P3) fails; if we understand it as 'true at Cosmo-Prime', (P2) does.

Prima facie this might look like a neat way to get around the argument for (C). Unfortunately, as we have already seen in chapter 1 (section 1.5), the claim that the truth schema holds in all contexts rules out this kind of context-sensitivity in monadic 'true'. For suppose 'true₁' and 'true₂' represented two admissible interpretations of 'true', we would have

The proposition that ϕ is true₁ iff ϕ

and

The proposition that ϕ is true₂ iff ϕ

and hence

The proposition that ϕ is true₁ iff it is true₂.

Thus, 'true' cannot be an independent source of context-sensitivity. If we want to say that 'the proposition that some_r swans are blue is true' has a hidden, truth-value-affecting contextual parameter whose value is a Lewis-world, we can do so only by saying the very same thing about 'some_r swans are blue'. But that would be to embrace the view that the relevant form of quantifier-restriction is just a kind of *de re* restriction.

The theorist who holds that some propositions are sets of Lewis-worlds is thus in the same position as the Relativists discussed in section 1.5. They have a relativised truth-predicate, to which they assign a certain theoretical importance; but they also admit a monadic truth-predicate, which conforms to the truth schema. They will thus accept (P2), which is an instance of the truth schema. Given that they want to accept (P1) while rejecting (C), they must thus simply reject (P3). Twin *does* say (and mean, and believe...) that some_r swans are blue; and what he believes is false, since no_r swans are blue. (P2'), however, is true: what Twin says is not false *at Cosmo-Prime*. *At Cosmo-Prime*, it is Twin who is right and I am who am wrong. The relation between Twin and me is like the relation between lovers and haters of Marmite, as

conceived by Relativists about ‘tasty’. The lovers believe that Marmite is tasty; the haters believe that it isn’t. Marmite either is or isn’t tasty, so either the lovers or the haters believe something false. But it’s not a big deal! For it is possible to believe something false without being “at fault”—not just in the epistemological sense that one’s belief is justified, or supported by one’s evidence, but in some deeper sense. The lovers and the haters both achieve ‘the goal of belief’, which requires only that the proposition believed be *true relative to the believer’s standards*, a status compatible with monadic falsehood.

As I discussed in chapter 1, the Relativist programme strikes me as extremely problematic even for ‘tasty’. According to Relativists, I ought to think that the things that are tasty for me are tasty, and the propositions that are true for me are true, while the propositions that are true for other people are often false; if they believe these propositions, as they ought, they will get things wrong. This seems to describe a state of extraordinary intellectual hubris: why should I be so confident that *I* am the special person truth for whom coincides with truth? The further claim that everyone else ought to (falsely) regard themselves as special in the same way does nothing to dispel the implausibility of this. However, at least in the case of ‘tasty’ there is a real philosophical itch that Relativism tries to scratch. It is certainly tempting to think those who, unlike me, dislike Marmite, *disagree* with me about whether Marmite is tasty, and thus believe (falsely) that it isn’t tasty. By reflecting on the difficulty of convincing the Marmite-haters of this, I could easily end up thinking that there is no good reasoning that could convince them that Marmite is tasty, and indeed, that good reasoning requires them to stick as firmly to their false belief as it requires me to stick to my true one. I can imagine getting this far and still feeling that in some sense we have no *deep* disagreement—and obscure as it is, the “no fault” idea perhaps does something to articulate this elusive notion of shallowness.

By contrast, when the difference between the relevant others and us is just that we are in different Lewis-worlds, there is nothing even *prima facie* appealing about Relativism. Why should I think, just on the grounds that Twin is not spatiotemporally related to me, that he will end up forming lots of false beliefs, for example the belief that some_r swans are blue, just by reasoning as he should? It is not at all plausible that, just because I think there are no swans in my Lewis-world are blue while Twin thinks that some swans in his Lewis-world are blue, Twin and I are committed to *disagreeing* about some proposition (viz. that some_r swans are blue). This idea is no more plausible than the idea that there is a proposition which

you and I are committed to disagreeing about whenever I think that there is beer in my house and you think that there is no beer in yours. One would only be tempted to think that Twin believes propositions inconsistent with propositions I believe if one confused this claim with some counterfactual claim to the effect that if things had been thus and so, I would have believed some propositions inconsistent with propositions I actually believe. But that would be a bad confusion: it would be like inferring the false conclusion that there is a city that Twin hates and I like from the (perhaps true) premise that there is a city I actually like but would have hated if I had instantiated Twin's qualitative profile.²²

The case for (C) stands. When people in other Lewis-worlds restrict their quantifiers in the way that leads to contingency, they generally do not express the same propositions that we express when we restrict our quantifiers in an analogous way. This is straightforwardly explained by the hypothesis that contingency-inducing quantification is *de re* restricted quantification. I grant that other explanations are conceivable. One could, for instance, propose that vim comes in a multitude of different flavours: our contingency-inducing quantifiers are restricted to things with our flavour of vim, while Twin's contingency-inducing quantifiers are restricted to things with his different flavour of vim.²³ But what is the point? Given the strength of the independent reasons for thinking that *de re* restricted quantification is pervasive, it is superfluous to postulate a new realm of properties, different in character both from ordinary qualitative properties and from *de re* properties, merely in order to provide an *additional* source of contingency in quantified claims. The best way to be a modal realist is to endorse BMCT, and learn to love the idea that all contingency is *de re* contingency.

²²Ironically, one of MacFarlane's favourite ways of defending Relativism in other domains is to appeal to something very like the view I have just been criticising: 'Consider Jane (who inhabits this world, the actual world) and June, her counterpart in another possible world. Jane asserts that Mars has two moons, and June denies this very proposition. Do they disagree? Not in any real way.' (MacFarlane 2007, p. 10)

²³Parsons (2002) considers a view that works rather like this in the temporal case. On the version of the A-theory he sets out, each instant of time has its own distinctive, intrinsic, monadic property, a 'degree of futurity or pastness'. These properties could have been distributed differently from how they in fact are distributed; for example, any given instant could have instantiated the particular property we call 'presentness', and indeed, each instant that does not instantiate this property either did once instantiate it, or will eventually instantiate it. Uses of the word 'present' at a given instant pick out whichever degree of futurity or pastness characterises that instant.

3.6 Necessary and eternal existence

Unrestrictedly speaking, everything is identical to something. *A fortiori*, every counterpart of anything is identical to something. Thus BMCT entails the very controversial claim that, unrestrictedly speaking, each thing is necessarily identical to something:

(NE) $\forall x \Box \exists y (y = x)$.

If we follow the logician's custom of understanding 'exists' as synonymous with 'is identical to something', we get the headline-grabbing claim that *each thing necessarily exists*. This is not a peculiarity of BMCT: since every counterpart of anything that is in a Lewis-world is identical to something in that Lewis-world, Lewis's analysis RCT also entails NE. Most philosophers have found this claim extremely implausible. According to them, Prince Charles, for example, could have failed to be identical to anything at all: Queen Elizabeth and Prince Philip could have failed to meet; if so, Prince Charles would never have been conceived; and in that case, there would have been nothing identical to him. This is one of the standard objections to counterpart theory.

A parallel objection could be raised to temporal counterpart theory. Just as BMCT entails NE, BTCT entails that each thing has always been, and always will be, identical to something (unrestrictedly speaking):

(EE) $\forall x H \exists y (y = x)$

$\forall x G \exists y (y = x)$

This is deeply controversial too. It is rejected, influentially, by 'presentists' in the tradition of Prior (1957, 1967, 1968b). However, temporal counterpart theorists who endorse EE have some influential allies, namely those philosophers—temporal B-theorists, for the most part—who take expressions like 'always' to be quantifiers that bind syntactically real variables ranging over times (see section 2.1). Those who think this must perforce take tense operators to be vacuous when applied to sentences that contain no variables for them to bind. And when the quantifier is read as unrestricted, it is hard to see how ' $\exists y (y = x)$ ' could contain any such variables. In the modal case, by contrast, NE is decidedly a minority view, although its popularity has grown somewhat in recent years thanks to the efforts of Zalta and Linsky (1994) and especially Timothy Williamson (1998, 2002, 2010, 2013).

Lewis's reaction to this objection to counterpart theory is quite concessive. He clearly thinks that there is a natural reading of 'Humphrey could have failed to exist' on which it is true iff there are Lewis-worlds that contain no counterparts of Humphrey. But he seems to despair of the task of providing a systematic semantic account of modality in English which explains how the sentence comes to have this reading:

What is the correct counterpart-theoretic interpretation of the modal formulas of the standard language of quantified modal logic? —Who cares? We can make them mean whatever we like. We are their master. We needn't be faithful to the meanings we learned at mother's knee—because we didn't. If this language of boxes and diamonds proves to be a clumsy instrument for talking about matters of essence and potentiality, let it go hang. Use the resources of modal realism *directly* to say what it would mean for Humphrey to be essentially human, or to exist contingently. (OPW 12–13)

I find this passage rather mystifying. True, I didn't learn about '□' and '◇' at mother's knee: but I *did* learn the words 'necessarily' and 'possibly' somewhere thereabouts, and later I was introduced to '□' and '◇' as synonyms for 'necessarily' and 'possibly'. True, the resources of English for talking about modal matters are rich. As well as 'necessarily' and 'possibly', we have more adverbs like 'essentially' and 'contingently', as well modal auxiliaries like 'can' and 'could', adjectives like 'possible' and 'potential', etc. We should be open to learning that not everything that can be said using these resources can be said using 'necessarily' and 'possibly'. But Lewis is not even sketching an approach to semantics of these aspects of English. Rather, it sounds like he is saying that we should be content with one-off accounts of particular sentences and give up on systematic semantics.

In any case, in the light of what we have learned in this chapter, the suggestion that the existence of Lewis-worlds containing no counterparts of Humphrey would be enough to vindicate a true reading of 'Humphrey could have failed to be identical to anything' in which the quantifier is *unrestricted* looks hopeless. That suggestion is inextricably tied to a conception of modality in which quantification over Lewis-worlds play a central role, and in which that role involves providing restrictions for quantifiers, even ones that are originally unrestricted. I have argued that this conception should be replaced by one conforming to *Modal Redundancy*. And once we make this replacement, the fact that there are Lewis-worlds containing no counter-

parts of Humphrey will look completely irrelevant to the truth of 'Humphrey could have failed to be identical to anything, unrestrictedly speaking', even if we are willing to entertain unorthodox ideas about the relation between this sentence and the language of boxes and diamonds.

I think a better option for modal counterpart theorists is to stand their ground and insist that it is just *true*, speaking unrestrictedly, that everything is necessarily identical to something. When unrestricted quantification is in play, modal realists are used to disagreeing with ordinary opinion. Nothing is more familiar to them than claiming, of some intuitively true sentence involving quantifiers, that it is true only when the quantifiers are taken to be somehow restricted. As we have seen, this is bound to happen with sentences like 'It is contingent whether some swans are blue'. Against the backdrop of this pattern of agreement and disagreement with pre-theoretical opinion, it is hard to see that counterpart theorists will incur any *further* costs if they claim, analogously, that 'It is contingent whether something is identical to Humphrey' is false when its quantifier is taken as unrestricted. So long as they agree that the sentence can be made true by interpreting its quantifier as restricted, the dialectical situation is just the familiar one.

And it is easy for counterpart theorists to come up with *restricted* readings of 'something' on which 'It is contingent whether something is identical to Humphrey' is true. For example, 'something' could carry a qualitative restriction, say to things that are *alive*. Then the sentence will be true so long as some but not all of Humphrey's counterparts are alive. Or 'something' could carry a *de re* restriction, say to things on Earth. While we have not yet considered how to analyse multiply *de re* modal claims like 'Possibly, nothing on Earth is identical to Humphrey'—this is a task for chapter 4—there is no reason at all to worry that they will turn out to be false. Or 'something' could carry a *de re* restriction to things in Cosmo. So long as our account of multiply *de re* modality can account for the truth of 'Possibly, nothing on Earth is identical to Humphrey', it will presumably be able to account in a parallel way for the truth of 'Possibly, nothing in Cosmo is identical to Humphrey.' None of these restricted readings of 'It is contingent whether Humphrey is identical to something' seems like a very plausible candidate to be what people actually have in mind when they utter sentences like 'It is contingent whether Humphrey exists'. The word 'exists' does not take so easily to contextual domain restriction as quantifiers proper: for example, I cannot felicitously say 'John does not exist' if all I mean is that John is not at my party. But I was not arguing that counterpart theorists should think of

themselves as agreeing with what the folk mean when they say that some things exist contingently; I was merely claiming that the disagreement, while real, is part of a familiar general pattern.

Lewis suggests that, in many ordinary contexts, 'counterpart' expresses a relation that human beings never bear to things that are not human, so that 'Each human being is necessarily a human being' is true. Unlike NE, this claim is not forced on us by the abstract form of the counterpart-theoretic analysis. And since 'counterpart' is certainly not one of the words we learnt at mother's knee, the right way to address the question whether human beings only human counterparts is not to directly consult our "speakers intuitions", but to consider the plausibility of the modal thesis that would follow from this claim. And *prima facie*, the claim that each human being is necessarily human looks rather implausible. Consider Prince Charles, for example. If we thought that it was necessary that he is a human being—an ordinary, flesh-and-blood human being, that is—we would have to say that he *would* have been human even if Queen Elizabeth and Prince Philip had never met (since obviously they *could* have failed to meet). That sounds implausible. Admittedly, our intuition that Charles would not have been human if Elizabeth and Philip had never met is intimately bound up with our temptation to think that there would have been nothing identical to Charles if Elizabeth and Philip had never met. So it is not clear how strongly counterpart theorists who resist that latter temptation should be swayed by the intuition. It is open to them to insist that Charles would still have been a flesh-and-blood human being even if Elizabeth and Philip had never met—perhaps a flesh-and-blood human being living on some planet that is not Earth, in some Lewis-world that is not Cosmo. But this radical claim is certainly not forced upon counterpart theorists. They could also say that Charles would under those circumstances have *failed* to be a flesh-and-blood human being, and indeed that he would have failed to be a physical thing with a mass, a shape, a location in space, etc. They could say, in other words, that Charles could have been *non-concrete*—that is, that some of his counterparts are non-concrete.

The idea that many concrete objects could have failed to be concrete plays an important role in Williamson's recent work defending NE. Williamson's idea is that ordinary claims about the possibility of things failing to exist are fairly reliable so long as we interpret 'exists' to mean something in the vicinity of 'is concrete', rather than 'is identical to something, unrestrictedly

speaking'. The exact meaning of 'concrete' is not so important: the general idea is that non-concrete things are things to which most ordinary, positive, non-modal predicates do not apply. I am inclined to think that proponents of BMCT should gratefully avail themselves of Williamson's suggestion. And as regards *temporal* counterpart theory, this seems absolutely crucial. The suggestion that if Elizabeth and Philip had never met, Charles would have been a flesh-and-blood human being living on a different planet was hard to take seriously; the analogous suggestion that *before* Elizabeth and Philip had ever met, Charles already was a flesh-and-blood human being (living on a different planet?) is just embarrassing.²⁴ But this is the sort of thing we would have to say if we claimed that it has always been the case that Charles is a human being, given that it has not always been the case that Elizabeth and Philip have met.

While "non-concrete" things are boring in many respects, they can have interesting modal and/or temporal properties. Charles has not always been a prince, but even when he was not a prince he was a *future* prince, and even when he has ceased to be a prince he will always be a *former* prince. By the lights of BTCT, this means that all Charles's past-counterparts, even the non-concrete ones, have princes among their future-counterparts, and all Charles's future-counterparts have princes among their past-counterparts. I, on the other hand, have never been a future prince, and will never be a past prince. My temporal counterparts, including the non-concrete ones, do not have princes among their temporal counterparts. Non-concrete objects can thus differ from one another as regards which objects they have as counterparts. This poses a challenge to the project of providing a non-circular analysis of temporal counterparthood in terms of something else: the relations we appeal to in the analysis must apply to non-concrete objects, and must make interesting distinctions between them. On Williamson's conception of non-concrete objects, this challenge is insuperable. On his picture, there are many distinctions among non-concrete objects that can *only* be captured in a language with temporal or modal operators, or their equivalents. However we should distinguish the general idea that before my birth and after my death I am still identical to something but lack mass, shape, a spatial location, a mental life, etc., from Williamson's own, particularly uncompromising, version of the view. What is needed to avoid absurdity is that a certain range of ordinary predicates (including 'overweight', 'in

²⁴I suppose that believes in the transmigration of souls might take this seriously. But they will not like the analogous claims about inanimate objects.

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